



SpendTogether

Track. Save. Grow. Together.

COMPLETE WEB APP SPECIFICATION

Income, Expense & Savings Web App

Product requirements · architecture · database · API · formulas
user flows · every screen · wireframes · design system
accessibility · responsive behaviour · testing · deployment

Next.js + Supabase

Multi-currency

Individuals & couples

Responsive PWA

DOCUMENT CONTROL

About this specification

Field	Value
Product	SpendTogether — Income, Expense & Savings web app (individuals and couples)
Document	Complete Web App Specification (PRD + architecture + design + delivery)
Version	1.0 — 17 September 2026
Platform	Responsive web app / installable PWA (desktop, tablet, mobile browsers)
Stack (confirmed)	Next.js (App Router, TypeScript, Node.js runtime) + Supabase (Postgres, Auth, Row-Level Security)
Currency model (confirmed)	Full multi-currency with conversion
Scope rule (confirmed)	Original PRD scope only. Multi-currency is the single approved scope exception.
Source documents	Income_Expense_Savings_App_Complete_PRD.pdf; Income_Expense_Savings_App_Mobile_Design_Specification.pdf; two SpendTogether design boards (PNG)
Owner	Corex Digital Solutions

How to read this document

The document is written for the people who will build, design, test and ship the product. Every requirement has an identifier so it can be traced into tickets and tests: **FR**- functional requirement, **BR**- business rule, **F**- formula, **SCR**- screen, **WAC**- web acceptance criterion, **T**- test case.

Normative language: **MUST** is mandatory for MVP; **SHOULD** is expected unless there is a documented reason; **MAY** is optional. Anything not stated here is out of scope for the MVP.

Scope interpretation rules

The owner chose to stay strictly inside the original PRD. That was applied as follows:

- **Added:** only multi-currency with conversion (explicitly approved).
- **Defined, not added:** the source documents reference behaviour without defining it (goal status thresholds, timezone and week boundaries, the Forgot Password screen, the DELETE /couple endpoint, category management). A spec with undefined behaviour cannot be built or tested, so this document gives each one the minimum definition needed. These are clarifications of existing scope, not new features.
- **Implementation endpoints:** a small number of API endpoints are required to implement screens the PRD already lists (profile/currency, categories, activity feed, exchange rates, password reset). They are listed separately in Section 10 and marked as such.
- **Excluded:** everything in the PRD non-goals, plus features the owner declined (goal withdrawals, recurring transactions, dark mode, budgeting). They are recorded in Section 26 as known limitations so nobody mistakes their absence for an oversight.

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READ FIRST

Source Corrections & Resolved Conflicts

The design boards are inspiration, not specification. They contain numerical and labelling errors. Where the boards conflict with the PRD or with arithmetic, this document wins. Every correction below is binding.

#	Source issue	Why it is wrong	Resolution in this spec
1	Dashboard hero shows "Available this month \$850" beside Income \$1,200, Expenses \$570, Saved \$300, Remaining \$330.	\$850 is not produced by any PRD formula. Net cash flow is 1,200 – 570 = 630; remaining cash flow is 1,200 – 570 – 300 = 330.	The hero card shows Remaining this month = Remaining Cash Flow (F-04). With the board data: \$330 . Net cash flow (\$630) is shown as a secondary metric.
2	Donut legend: Food 30%, Bills 25%, Transport 20%, Shopping 15%, Other 10% of \$570.	Dashboard shows Food \$140, Transport \$90, Bills \$150. 140/570 = 24.6%, 90/570 = 15.8%, 150/570 = 26.3%. The legend is invented.	Category % always computed with F-11. Reference data used throughout this spec: Food \$140 (24.6%), Bills \$150 (26.3%), Transport \$90 (15.8%), Shopping \$85 (14.9%), Other \$105 (18.4%). Total \$570, sums to 100%.
3	Board 1 typography: Inter Bold headings. Board 2: Plus Jakarta Sans.	Two competing type systems.	Plus Jakarta Sans for headings; Inter for body, UI and all numbers (tabular figures). Section 17.
4	Board 2 palette: "Accent #F59E0B (Amber)" swatch is purple; "Secondary #6366F1" swatch is grey; several labels garbled.	Labels and swatches disagree.	Hex values win: Primary #10B981, Secondary #6366F1, Accent #F59E0B, Success #22C55E, Error #EF4444, Neutral #6B7280 / #111827.
5	Primary #10B981 used for white button text and green text; amber and #22C55E used as text colours.	WCAG contrast: #10B981 on white = 2.54:1, #22C55E = 2.28:1, #F59E0B = 2.15:1. All fail the 4.5:1 text minimum.	Brand hues are kept for fills and illustration only. Text and filled buttons use accessible shades (Primary-700 #047857 = 5.48:1). Section 17.
6	Mobile boards show 4 tabs + central "+" (no Profile tab). Design spec defines 5 tabs (incl. Profile) + global Add.	Navigation conflict.	Written design spec wins: 5 destinations (Home, Activity, Goals, Insights, Profile) + a global Add action. Section 12.
7	PRD required-pace formula: Required Daily = Remaining ÷ Days Remaining, with Days Remaining = max(Target – Today, 0).	Divides by zero on the target date itself.	F-17 uses max(Days Remaining, 1) while the target date is today or later; after the date passes, required pace is "Overdue" (no number).
8	Board status chip "At Risk" and goal statuses referenced without thresholds ("configurable thresholds").	Untestable.	Concrete default thresholds defined in F-19 and stored in app_config so they remain configurable.

Decision

Mockup screens in this document use the corrected reference dataset. Developers **MUST NOT** copy numbers from the original boards into fixtures or tests.

PART A — PRODUCT

Product Requirements

1. Product overview

SpendTogether is a mobile-first, responsive web application for tracking income and expenses and for reaching savings goals, alone or as a couple. It addresses two problems: understanding where money comes from and where it goes, and helping users consistently reach individual or shared savings goals.

Core promise: Know what you earn. Know what you spend. Know what you can save.

Tagline: Track. Save. Grow. Together.

MVP scope: income tracking, expense tracking, daily/weekly/monthly insights, individual goals, couple goals, goal contributions, partner invitations, financial summaries, and multi-currency entry with conversion.

2. Goals and non-goals

Goals

- Record income and expenses in any supported currency, with categories, dates and optional notes.
- Review financial performance by day, week and month in the user's base currency.
- Calculate net cash flow, recorded savings, remaining cash flow and savings rate.
- Create individual and couple goals, contribute to them, and see whether each goal is on track.
- Keep personal finances private by default, including from a connected partner.

Explicit MVP non-goals (from PRD, unchanged)

Bank integrations, automatic transaction imports, payment processing, investments, debt/loan management, credit scoring, receipt scanning, AI advice, shared wallets, expense settlements, complex accounting, tax tools and advanced budgeting.

3. Users and product principles

Persona	Needs	Key screens
Individual	Simple visibility into income, spending, remaining money and personal goals. Often paid or spending in more than one currency.	Home, Add, Activity, Insights, Goals
Couple (two independent users)	Collaborate on shared savings without exposing personal transactions to each other.	Couple, Goals → Our Goals, Goal Details

1. **Fast entry beats financial complexity.** Adding a transaction takes one screen and under 10 seconds for a returning user.
2. **Personal finances remain private by default.** Enforced in the database, not only in the UI.
3. **Savings contributions are distinct from expenses.** They never appear in expense analytics.
4. **The app explains numbers, not merely stores them.** Every metric has a label and, where useful, a one-line explanation.
5. **Mobile-first and usable in a few taps,** while making good use of larger screens.

4. Core concepts and business rules

ID	Rule
BR-01	Income is money received by a user. Expense is money spent by a user. Both are <i>transactions</i> owned by exactly one user.
BR-02	Recorded savings is money explicitly contributed to a savings goal. A contribution is a savings event, never an expense, and MUST NOT appear in expense totals, category breakdowns or average spending.
BR-03	An individual goal is owned and funded by one user and has no couple reference.
BR-04	A couple goal belongs to an active couple. Either member may contribute; every contribution retains its contributor.
BR-05	Privacy rule: connecting as a couple does not expose either partner's income, expenses, balances, individual goals or personal summaries.
BR-06	A user belongs to at most one active or pending couple at a time. A couple has exactly two members when active.
BR-07	All monetary amounts are positive. Direction comes from type (income / expense / contribution), never from the sign of the amount.
BR-08	Goal balances are derived from contribution records. No mutable balance column exists.
BR-09	Transaction and contribution dates may not be later than "today" in the user's timezone.
BR-10	A goal target date must be today or later at creation. A goal target amount must be > 0.
BR-11	A goal is Completed automatically when its balance $\geq$ its target, and returns to an active status if a later edit or deletion drops the balance below target.
BR-12	Base currency: each user has one base (default) currency. All of that user's summaries, insights and savings-rate figures are expressed in it.
BR-13	Entry currency: any transaction or contribution may be entered in any supported currency. The original amount and currency are always preserved.
BR-14	Rate locking: conversion uses the stored daily rate for the record's date. Once stored, converted values do not change when market rates move (see Section 11 for the base-currency-change exception).
BR-15	Goal currency: each goal has one currency, set at creation (default: creator's base currency) and immutable afterwards. Contributions are converted into it.
BR-16	Periods are calendar periods in the user's IANA timezone. Day = local calendar day. Week = ISO week, Monday to Sunday. Month = calendar month.
BR-17	Categories: default categories are system-owned and cannot be edited or deleted. Users may create custom categories. A custom category in use is archived (hidden from pickers) instead of deleted.
BR-18	Couple end: DELETE /couple (a PRD endpoint) ends the couple. Couple goals become read-only for both former partners; no new contributions; history remains visible to both. Nothing else is shared or moved.

Note

BR-16 and BR-18 are the minimum definitions required for PRD behaviour that the PRD references but never defines (period boundaries; the DELETE /couple endpoint). They are not new features.

5. Functional Requirements

Grouped by capability. Each requirement maps to at least one acceptance criterion in Section 22 and one test in Section 23.

ID	Requirement	Priority	PRD ref
FR-01	Register with name, unique email or phone, password and confirm password. Duplicate identifiers are rejected with a non-revealing message.	MUST	AC01
FR-02	Log in with identifier + password. Session persists across reloads via HTTP-only secure cookies.	MUST	AC01
FR-03	Log out ends the session on the current device.	MUST	\$10
FR-04	Forgot / reset password via emailed link (email accounts) or SMS code (phone accounts).	MUST	Design \$3, \$7
FR-05	First-run setup: choose base currency (pre-selected from browser locale) and confirm timezone (auto-detected).	MUST	\$5 Auth flow
FR-06	Add income: amount, currency (default base), category, date (default today), optional note.	MUST	AC02
FR-07	Add expense: same fields; recent categories listed first.	MUST	AC03
FR-08	Edit and delete any own transaction; all affected summaries recalculate.	MUST	AC02, AC03
FR-09	Activity list grouped by date, with filters (All, Income, Expense, Savings Contributions, category, date range) and text search on note/category.	MUST	\$9
FR-10	Home dashboard for Today / This Week / This Month: income, expenses, recorded savings, remaining cash flow, net cash flow, savings rate, category preview, active goals, quick add.	MUST	AC04
FR-11	Insights for Daily / Weekly / Monthly: summary metrics, spending trend, income vs expenses, category breakdown, savings rate, previous-period comparison.	MUST	AC13
FR-12	Create individual goal: name, target amount, goal currency, target date.	MUST	AC06
FR-13	Create couple goal only when an active couple exists; otherwise the option is disabled with an explanation.	MUST	AC07
FR-14	Add contribution to a goal: amount, currency (default goal currency), date, optional note.	MUST	AC08
FR-15	Edit/delete own contributions (needed for AC08 derived balances to remain correct after mistakes).	MUST	AC08, AC10
FR-16	Goal details: target, balance, remaining, progress, required pace, current pace, projected completion, status, contributor breakdown (couple), history.	MUST	AC09
FR-17	Goal auto-completes when balance $\geq$ target (BR-11).	MUST	AC10
FR-18	Edit goal name, target amount and target date; delete goal (with confirmation).	MUST	\$10 PATCH/DELETE /goals
FR-19	Invite partner by email or phone; invitee registers or logs in, then accepts. Inviter can cancel or resend a pending invitation.	MUST	AC11
FR-20	Couple screen shows state (none / pending / connected), partner identity and shared goals; never private data.	MUST	AC12
FR-21	End couple (DELETE /couple) with a confirmation explaining consequences (BR-18).	MUST	\$10
FR-22	Profile & settings: name, email/phone, base currency, timezone, categories, notifications, couple controls, logout.	MUST	\$9
FR-23	Manage custom categories (create, rename, change icon, archive).	MUST	\$9 Profile
FR-24	Enter amounts in any supported currency; show the converted base-currency value before saving.	MUST	Scope exception

ID	Requirement	Priority	PRD ref
FR-25	Offline: queue transaction and contribution creation locally and sync automatically, with a "Sync pending" indicator.	SHOULD	Design §21
FR-26	Email notifications: partner invitation, invitation accepted, couple goal completed. User can switch off the last two.	SHOULD	§9 Notifications

DOMAIN LOGIC

6. Financial Formulas

All formulas live in one pure TypeScript package (@spendtogether/domain) with no framework, database or UI imports. The API, server components and offline client all call the same functions. This is the PRD requirement that "financial calculations live in reusable domain services" and are unit-tested independently of the UI.

6.1 Money representation

- Amounts are stored as **integers in minor units** (bigint), e.g. \$12.50 = 1250 cents. Never floating point.
- Each currency carries its ISO 4217 exponent (USD 2, LRD 2, JPY 0, KWD 3). Minor units = major $\times 10^{\text{exponent}}$.
- Exchange rates are stored as `numeric(24,10)` and multiplied using a decimal library (decimal.js) on the server.
- Conversion rounds **half away from zero** to the target currency's minor unit, once per record. Aggregates sum already-rounded converted values; they are never re-converted.
- Percentages are computed in full precision and rounded to one decimal place only for display.

6.2 Period calculations (base currency)

All sums use each record's converted base amount (base_amount_minor) and include only records whose date falls in the selected period (BR-16).

ID	Name	Formula	Notes
F-01	Total Income	$\Sigma \text{base_amount}$ of income transactions in period	
F-02	Total Expenses	$\Sigma \text{base_amount}$ of expense transactions in period	Excludes contributions (BR-02).
F-03	Recorded Savings	$\Sigma \text{contributor_base_amount}$ of the user's own contributions in period	Includes the user's contributions to couple goals, excludes the partner's.
F-04	Remaining Cash Flow	$F-01 - F-02 - F-03$	May be negative; shown as "Overspent" with a minus sign and icon.
F-05	Net Cash Flow	$F-01 - F-02$	May be negative.
F-06	Savings Rate	$(F-03 \div F-01) \times 100$	If $F-01 = 0 \rightarrow$ "N/A" (AC05). Can exceed 100% (saving from earlier income); displayed as computed.
F-07	Average Daily Spending	$F-02 \div D$	D = calendar days in the period. For the current, unfinished period, D = days elapsed including today. Always ≥ 1 .
F-08	Category Total	$\Sigma \text{expense base_amount}$ in category in period	
F-09	Category %	$(F-08 \div F-02) \times 100$	If $F-02 = 0 \rightarrow$ no breakdown shown (empty state).
F-10	Period Change	$(\text{Current} - \text{Previous}) \div \text{Previous} \times 100$	If Previous = 0 $\rightarrow$ show absolute change only, label "New".

Rule

F-07 clarification: dividing the current month's spending by 30 on the 3rd of the month would understate daily spending tenfold. For completed periods D is the full calendar length, exactly as the PRD states; for the current period D counts elapsed days only.

6.3 Goal calculations (goal currency)

ID	Name	Formula
F-11	Goal Balance	Σ goal_amount_minor of all contributions to the goal
F-12	Remaining Amount	$\max(\text{Target} - \text{F-11}, 0)$
F-13	Progress %	$\min((\text{F-11} \div \text{Target}) \times 100, 100)$
F-14	Days Remaining	$\max(\text{TargetDate} - \text{Today}, 0)$ in whole days, user timezone
F-15	Required Daily	If TargetDate $\geq$ Today: $\text{F-12} \div \max(\text{F-14}, 1)$. If TargetDate < Today and not completed: "Overdue" (no value).
F-16	Required Weekly	$\text{F-15} \times 7$
F-17	Required Monthly	$\text{F-15} \times 30.4375$
F-18	Current Pace (daily)	Σ contributions in the last 30 days $\div$ $\min(30, \text{goal age in days}, \geq 1)$
F-19	Status	See 6.4
F-20	Projected Completion	If $\text{F-18} > 0$: Today + $\text{ceil}(\text{F-12} \div \text{F-18})$ days. Else "No recent contributions".
F-21	Contributor Share (couple)	Σ contributor's goal_amount $\div$ $\text{F-11} \times 100$

6.4 Goal status (F-19)

The PRD asks to compare actual contribution pace with required pace using configurable thresholds. The status uses the expected-balance method, which compares actual progress against a straight-line plan from creation to target date:

```

elapsed      = max(Today - CreatedDate, 0)      -- days
total        = max(TargetDate - CreatedDate, 1)  -- days
expected     = Target * min(elapsed / total, 1)
pace_ratio   = expected == 0 ? 1 : Balance / expected

if Balance >= Target      -> COMPLETED
else if Today > TargetDate -> BEHIND (overdue)
else if pace_ratio >= ON_TRACK_MIN (0.95) -> ON_TRACK
else if pace_ratio >= AT_RISK_MIN (0.75)  -> AT_RISK
else                          -> BEHIND

```

- Thresholds ON_TRACK_MIN and AT_RISK_MIN live in the app_config table, so they are configurable without a deploy.
- A brand-new goal has expected = 0 and is On Track, so users are not told they are behind on day one.
- Status is always shown as text + icon, never colour alone.

6.5 Worked example (reference dataset)

Input / output	Value
Income (month)	\$1,200.00
Expenses (month)	\$570.00
Recorded savings (month)	\$300.00
Net cash flow (F-05)	\$630.00
Remaining cash flow (F-04)	\$330.00
Savings rate (F-06)	25.0%
Avg daily spending on 17 Sep (F-07)	$\$570 \div 17 = \33.53
Goal "New Laptop": target \$1,200, balance \$600, created 1 Jul, due 31 Dec	Progress 50.0%; remaining \$600; 105 days left → required \$5.71/day, \$40.00/week, \$173.93/month

Input / output	Value
Status check	elapsed 78 of 183 days → expected \$511.48 → ratio 1.17 → On Track

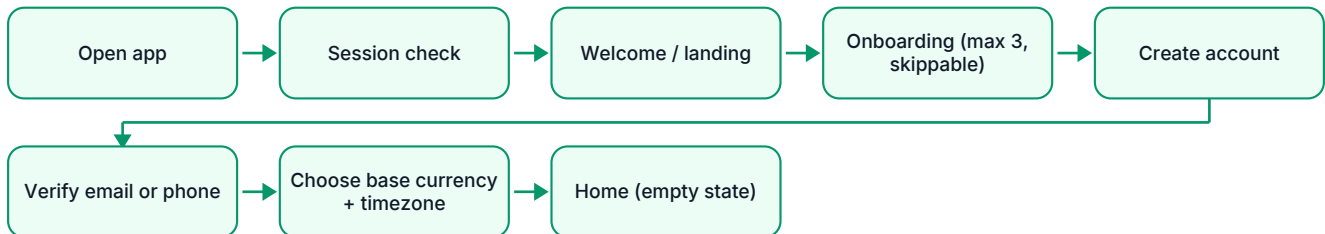
6.6 Currency formulas

ID	Name	Formula
F-22	Cross rate A→B on date d	$\text{rate}(\text{USD} \rightarrow \text{B}, d) \div \text{rate}(\text{USD} \rightarrow \text{A}, d)$; A = B → 1
F-23	Converted amount	$\text{round_half_away}(\text{amount_minor_A} \div 10^{\text{expA}} \times \text{rate}(\text{A} \rightarrow \text{B}) \times 10^{\text{expB}})$
F-24	Rate for date d	Latest stored rate with $\text{rate_date} \leq d$. If none exists (before launch), earliest stored rate, flagged $\text{fx_estimated} = \text{true}$.

7. Complete User Flows

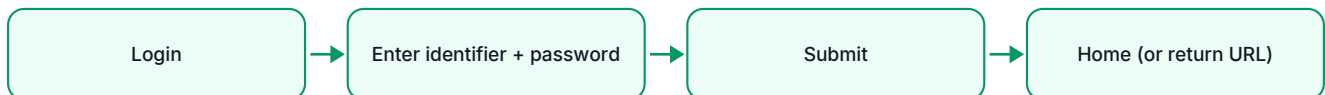
Each flow lists the happy path as a diagram, followed by branches. "Toast" means a non-blocking confirmation message. Optimistic updates are applied immediately and rolled back on failure.

7.1 First launch, registration and setup



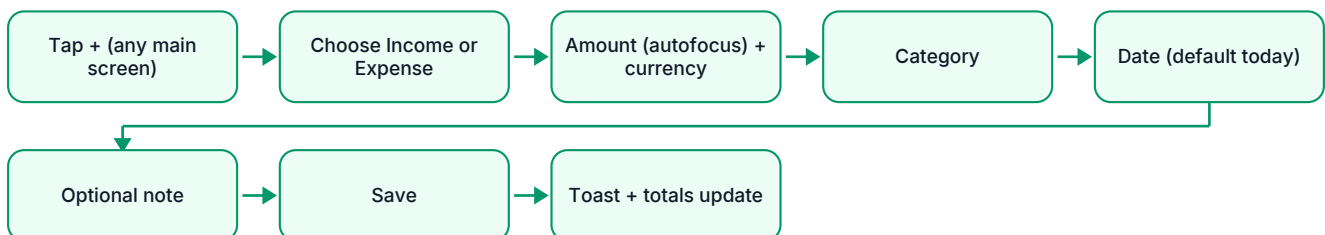
- Valid session found → skip to Home.
- Duplicate identifier → field error: "An account with these details already exists. Log in instead?" (no disclosure beyond that).
- Verification link expired → resend link screen.

7.2 Login and password reset



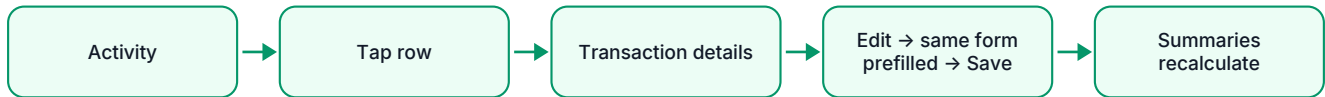
- Wrong credentials → "Email/phone or password is incorrect." Rate limited after 5 failures per 15 min.
- Forgot password → enter identifier → email link or SMS code → set new password → all other sessions revoked → Login.

7.3 Add income / add expense



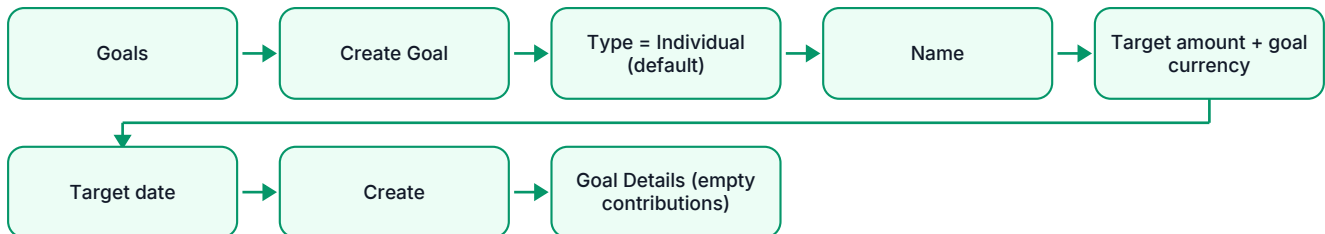
- Currency ≠ base → live preview "≈ \$X.XX in USD at today's rate".
- Offline → saved to queue, row shows "Sync pending".
- Server error → form stays open with values preserved + Retry.

7.4 Edit or delete a transaction



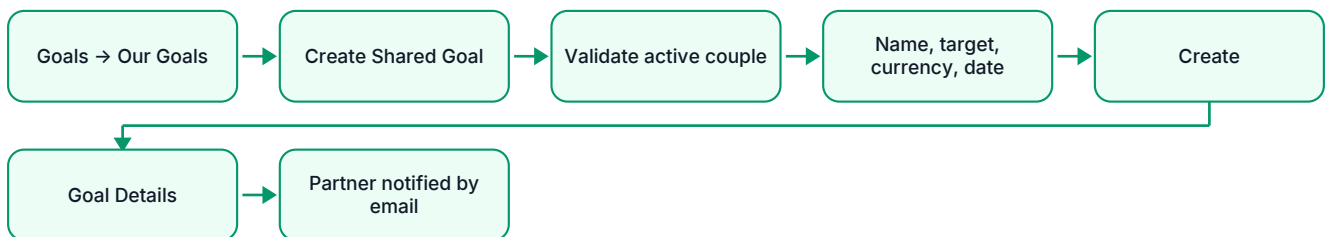
- Delete → dialog "Delete this expense? Your totals and insights will be recalculated." → Confirm → return to Activity → Toast with Undo (5 s, soft-delete).

7.5 Create individual goal



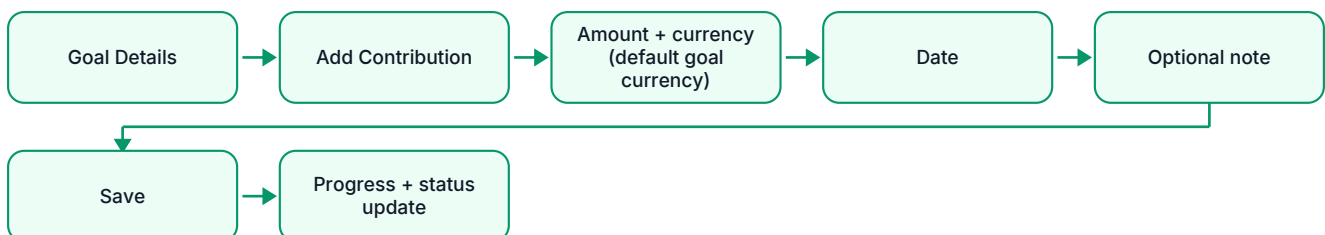
- Target ≤ 0 or date in past → inline error; Create stays disabled.

7.6 Create couple goal



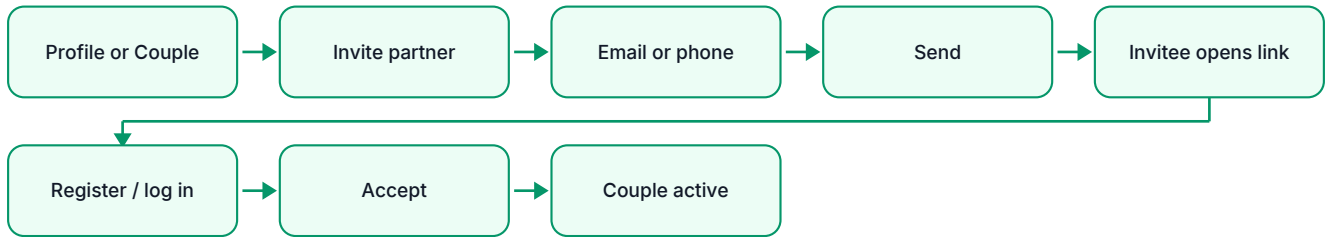
- No active couple → Couple option disabled with text "Connect a partner to create shared goals" + Connect Partner link.

7.7 Add goal contribution



- Balance reaches target → Completed state with restrained celebration; partner emailed for couple goals.
- Contribution in other currency → preview "≈ €X in goal currency".

7.8 Invite and connect partner

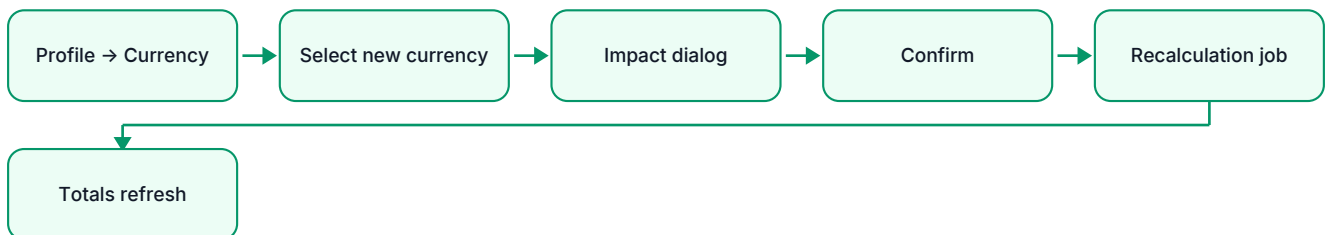


- Invitee already in a couple → "This person can't accept right now." Inviter sees invitation expire.
- Invitation expires after 7 days → Resend available.
- Inviter cancels → token invalidated immediately.
- Invitee declines → inviter sees "Declined".

7.9 End couple



7.10 Change base currency



- During recalculation, dashboards show a banner "Updating your totals to EUR..." and use the previous values.

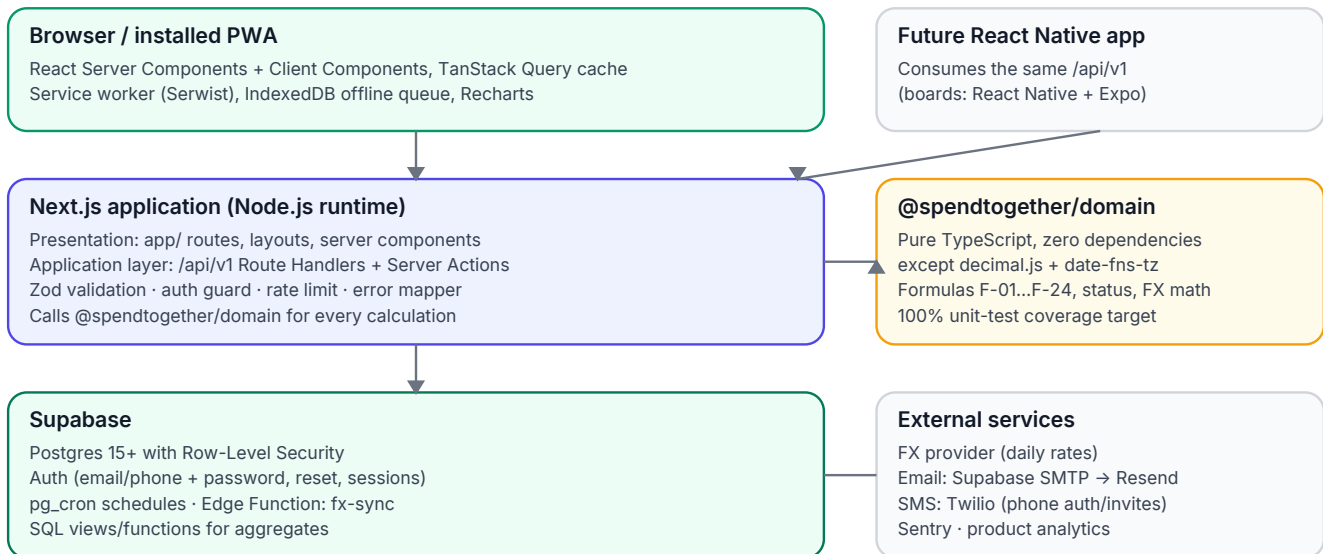
PART B — ARCHITECTURE

8. Product Architecture

8.1 Stack decision

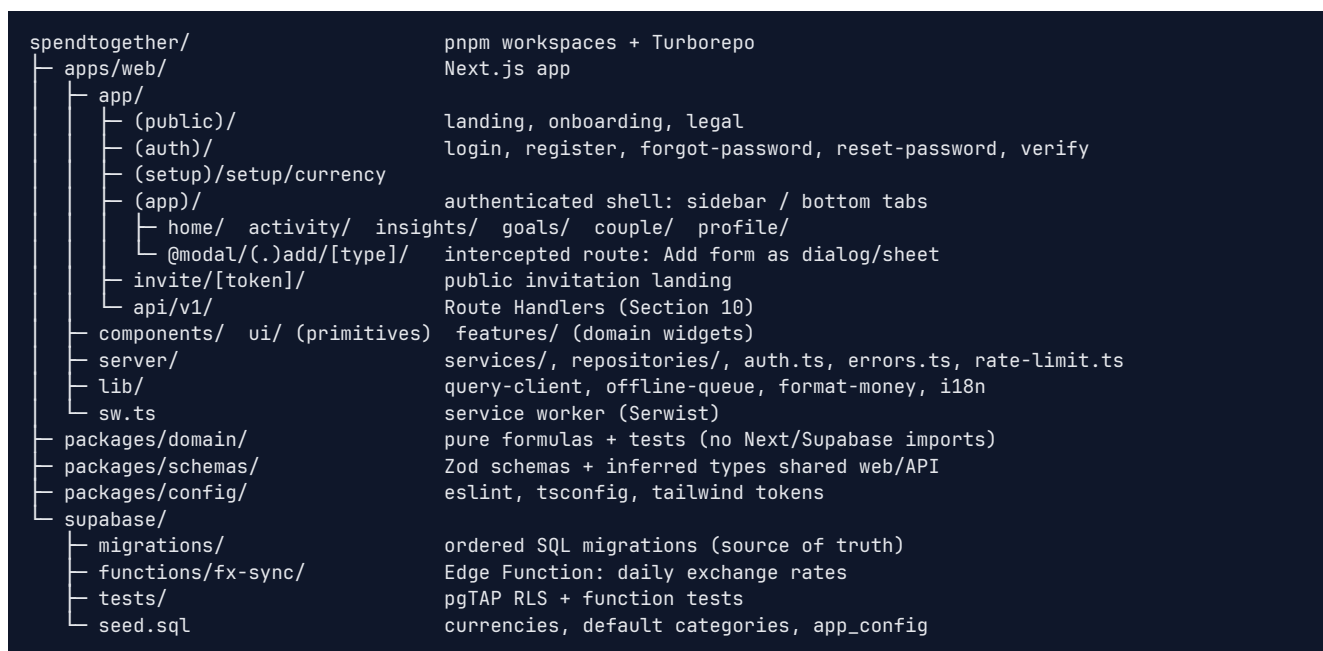
Layer	Choice	Reason
Framework	Next.js App Router, TypeScript strict, Node.js runtime	One codebase for UI and API; server components keep financial queries on the server; matches team skills.
API	Route Handlers under /api/v1 (REST)	Implements the PRD endpoint list exactly; reusable by a later React Native client.
UI mutations (web)	Server Actions calling the same application services	Fewer round trips for forms; one service layer for both paths.
Database	Supabase Postgres	Relational model required by PRD §7; RLS enforces privacy at the data layer.
Auth	Supabase Auth via @supabase/ssr (HTTP-only cookies)	Hashing, sessions, refresh, reset built in. Satisfies "tokens not in plain application storage".
Data access	Supabase client (typed with generated types) + SQL functions for aggregates	Types generated from the schema; heavy sums done in Postgres, not JavaScript loops.
Validation	Zod schemas shared by forms and API	One definition of "valid" on client and server; server validation is authoritative.
Client state	TanStack Query (cache, optimistic updates, persistence to IndexedDB)	Instant-feeling interactions and offline reads.
Styling	Tailwind CSS with design tokens as CSS variables; Radix UI primitives for dialogs, sheets, menus	Accessible primitives; tokens map 1:1 to Section 18.
Charts	Recharts	Composable React charts; accessible text tables added alongside (Section 16).
PWA/offline	Servist service worker + IndexedDB (idb)	Offline queue and installability.
Testing	Vitest, pgTAP (supabase test db), Playwright, axe-core, Lighthouse CI	Section 23.

8.2 System context



Layering (PRD §12): presentation → API/application → domain → data access → relational database. Dependencies only point downward.

8.3 Repository structure



8.4 Request lifecycle (example: add expense)

1. Client validates with the shared Zod schema and applies an optimistic row + totals via TanStack Query.
2. POST /api/v1/transactions with an Idempotency-Key header (client-generated UUID).
3. Route Handler: resolve session from cookie → rate limit → validate body → call `TransactionService.create()`.
4. Service loads the user's base currency and the FX rate for the date, calls `domain.convert()`, writes the row through the user-scoped Supabase client (RLS active).
5. Response returns the stored transaction; client invalidates summary, insights, activity query keys for affected periods.

8.5 Rendering strategy

Surface	Rendering	Caching
Landing, onboarding, legal	Static (SSG)	CDN
App shell (nav, header)	Server component, per-request	No shared cache (personal data)
Home, Insights, Goals	Server component for first paint with streamed Suspense sections; client hydration takes over with TanStack Query	Query staleTime 30 s; revalidated on focus and after mutations
Forms (add/edit)	Client components	n/a
Invitation landing	Server component; token verified server-side	No cache

Rule

The service-role Supabase key is used ONLY in server-side jobs (fx-sync, base-currency recalculation, invitation acceptance function). All user-initiated reads and writes go through the user-scoped client so RLS always applies.

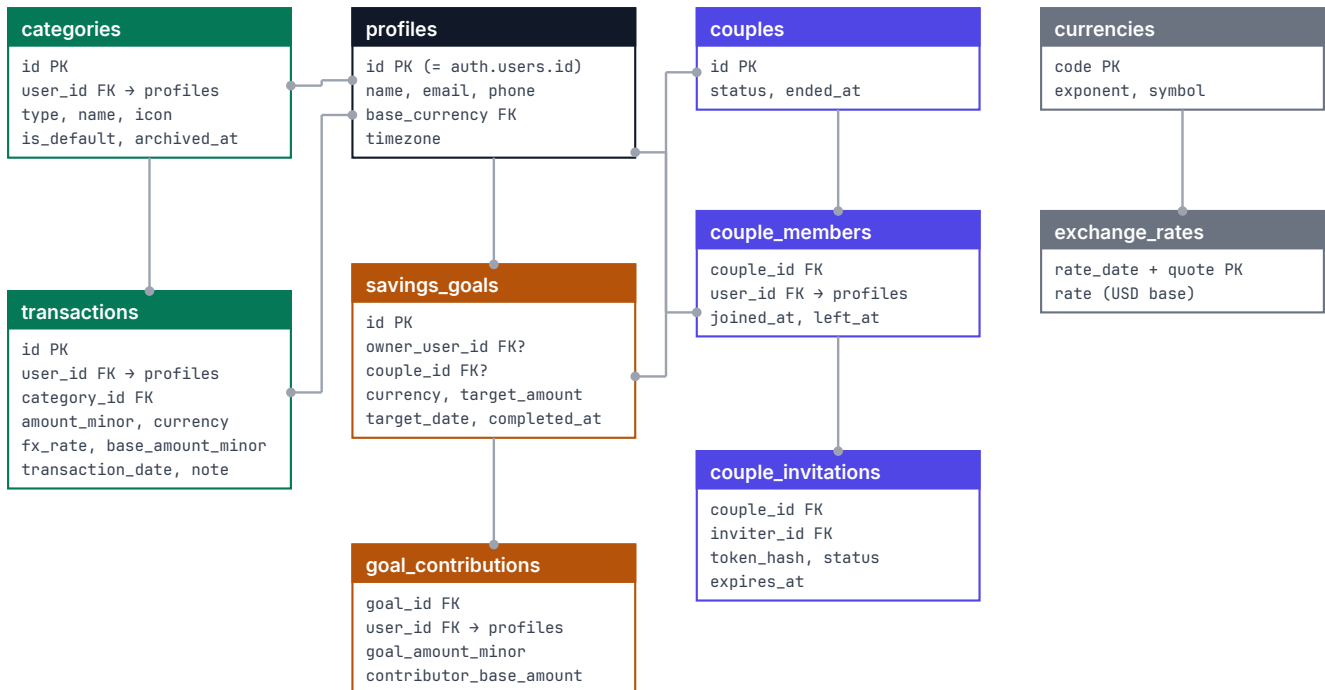
9. Database & Schema

Postgres on Supabase. Migrations in `supabase/migrations` are the single source of truth; TypeScript types are generated from them in CI. The model follows PRD §7 with three extensions: currency fields (approved scope exception), soft delete for Undo (design spec §13), and supporting tables for FX rates, idempotency and configuration.

9.1 Entity overview

Entity	PRD purpose	Key fields (this spec)
profiles	Account owner ("users")	id, name, email, phone, base_currency, timezone, timestamps
couples	Couple relationship	id, status (pending/active/ended), ended_at
couple_members	Connects two users	couple_id, user_id, joined_at, left_at
couple_invitations	Invitation lifecycle	couple_id, inviter_id, invitee, token_hash, status, expires_at
categories	Income/expense categories	user_id (nullable), name, type, icon, color, is_default, archived_at
transactions	Income and expenses	user_id, type, amount_minor, currency, fx_rate, base_amount_minor, category_id, transaction_date, note
savings_goals	Individual/couple targets	type, owner_user_id or couple_id, name, target_amount_minor, currency, target_date, completed_at
goal_contributions	Goal funding history	goal_id, user_id, amount_minor, currency, goal_amount_minor, contributor_base_amount_minor, contribution_date
exchange_rates	(multi-currency)	rate_date, quote, rate (USD base)
currencies	(multi-currency)	code, name, symbol, exponent
idempotency_keys / app_config	(infrastructure)	offline-safe retries; thresholds and settings

9.2 Entity-relationship diagram



Dots mark relationship ends. All currency columns (profiles, transactions, goals, contributions) reference currencies.code; lines omitted for clarity.

9.3 DDL

```

-- 0001_core.sql (abridged: comments, triggers for updated_at omitted)
create extension if not exists pgcrypto;

create table currencies (
  code      char(3) primary key,          -- ISO 4217
  name      text not null,
  symbol    text not null,
  exponent  smallint not null check (exponent between 0 and 4),
  is_active boolean not null default true
);

create table profiles (
  id        uuid primary key references auth.users(id) on delete cascade,
  name      text not null check (char_length(name) between 1 and 80),
  email     citext unique,
  phone     text unique,                  -- E.164
  base_currency char(3) not null references currencies(code),
  timezone  text not null default 'UTC',  -- IANA
  notify_email jsonb not null default '{"invite_accepted":true,"goal_completed":true}',
  onboarded_at timestamptz,
  created_at timestamptz not null default now(),
  updated_at timestamptz not null default now(),
  check (email is not null or phone is not null)
); -- password_hash lives in auth.users (Supabase Auth, bcrypt)

create type couple_status as enum ('pending','active','ended');
create table couples (
  id        uuid primary key default gen_random_uuid(),
  status    couple_status not null default 'pending',
  created_at timestamptz not null default now(),
  ended_at  timestamptz,
  updated_at timestamptz not null default now()
)
  
```

```

);

create table couple_members (
  id          uuid primary key default gen_random_uuid(),
  couple_id   uuid not null references couples(id) on delete cascade,
  user_id     uuid not null references profiles(id) on delete cascade,
  joined_at   timestampz not null default now(),
  left_at     timestampz,
  unique (couple_id, user_id)
);
-- BR-06: at most one open membership per user
create unique index one_open_couple_per_user on couple_members(user_id) where left_at is null;

create type invitation_status as enum ('pending','accepted','declined','cancelled','expired');
create table couple_invitations (
  id          uuid primary key default gen_random_uuid(),
  couple_id   uuid not null references couples(id) on delete cascade,
  inviter_id  uuid not null references profiles(id) on delete cascade,
  invitee     text not null, -- normalised email or E.164 phone
  invitee_kind text not null check (invitee_kind in ('email','phone')),
  token_hash  text not null unique, -- sha256 of token; raw token only in the link
  status      invitation_status not null default 'pending',
  expires_at  timestampz not null default now() + interval '7 days',
  responded_at timestampz,
  created_at  timestampz not null default now()
);
create unique index one_pending_invite_per_couple on couple_invitations(couple_id) where status = 'pending';

create type category_type as enum ('income','expense');
create table categories (
  id          uuid primary key default gen_random_uuid(),
  user_id     uuid references profiles(id) on delete cascade, -- null = default/system
  name        text not null check (char_length(name) between 1 and 40),
  type        category_type not null,
  icon        text not null, -- lucide icon key
  color       text not null, -- chart token key, e.g. 'cat-food'
  is_default  boolean not null default false,
  archived_at timestampz,
  created_at  timestampz not null default now(),
  check ((is_default and user_id is null) or (not is_default and user_id is not null))
);
create unique index categories_unique_name on categories(coalesce(user_id,'00000000-0000-0000-0000-000000000000'), type, lower(name));

create type transaction_type as enum ('income','expense');
create table transactions (
  id          uuid primary key default gen_random_uuid(), -- client may supply (offline)
  user_id     uuid not null references profiles(id) on delete cascade,
  type        transaction_type not null,
  amount_minor bigint not null check (amount_minor > 0),
  currency    char(3) not null references currencies(code),
  fx_rate     numeric(24,10) not null check (fx_rate > 0), -- currency -> base
  fx_rate_date date not null,
  fx_estimated boolean not null default false,
  base_currency char(3) not null references currencies(code),
  base_amount_minor bigint not null check (base_amount_minor > 0),
  category_id uuid not null references categories(id),
  transaction_date date not null,
  note        text check (char_length(note) <= 280),
  deleted_at  timestampz, -- soft delete for Undo
  created_at  timestampz not null default now(),
  updated_at  timestampz not null default now()
);
create index tx_user_date on transactions(user_id, transaction_date desc) where deleted_at is null;
create index tx_user_type_date on transactions(user_id, type, transaction_date) where deleted_at is null;
create index tx_user_cat on transactions(user_id, category_id) where deleted_at is null;

create type goal_type as enum ('individual','couple');
create table savings_goals (
  id          uuid primary key default gen_random_uuid(),
  type        goal_type not null,
  owner_user_id uuid references profiles(id) on delete cascade,
  couple_id   uuid references couples(id) on delete cascade,
  created_by  uuid not null references profiles(id),
  name        text not null check (char_length(name) between 1 and 60),
  icon        text not null default 'target',
  target_amount_minor bigint not null check (target_amount_minor > 0),
  currency    char(3) not null references currencies(code), -- BR-15, immutable
  target_date date not null,
  completed_at timestampz,
  archived_at timestampz, -- set when couple ends

```

```

    created_at      timestampz not null default now(),
    updated_at      timestampz not null default now(),
    check ((type='individual' and owner_user_id is not null and couple_id is null)
           or (type='couple' and couple_id is not null and owner_user_id is null))
);

create table goal_contributions (
    id              uuid primary key default gen_random_uuid(),
    goal_id          uuid not null references savings_goals(id) on delete cascade,
    user_id          uuid not null references profiles(id), -- contributor
    amount_minor     bigint not null check (amount_minor > 0),
    currency         char(3) not null references currencies(code),
    goal_fx_rate      numeric(24,10) not null,
    goal_amount_minor bigint not null check (goal_amount_minor > 0),
    contributor_base_currency char(3) not null references currencies(code),
    contributor_fx_rate numeric(24,10) not null,
    contributor_base_amount_minor bigint not null check (contributor_base_amount_minor > 0),
    fx_rate_date      date not null,
    fx_estimated      boolean not null default false,
    contribution_date date not null,
    note             text check (char_length(note) <= 280),
    created_at        timestampz not null default now(),
    updated_at        timestampz not null default now()
);

create index contrib_goal_date on goal_contributions(goal_id, contribution_date desc);
create index contrib_user_date on goal_contributions(user_id, contribution_date);

create table exchange_rates (
    -- USD-based daily snapshot
    rate_date    date not null,
    quote        char(3) not null references currencies(code),
    rate         numeric(24,10) not null check (rate > 0), -- 1 USD = rate QUOTE
    source       text not null,
    fetched_at    timestampz not null default now(),
    primary key (rate_date, quote)
);

create table idempotency_keys (
    key          uuid not null,
    user_id      uuid not null references profiles(id) on delete cascade,
    endpoint     text not null,
    response     jsonb not null,
    created_at   timestampz not null default now(),
    primary key (user_id, key)
); -- purged after 48 h by pg_cron

create table app_config (
    key  text primary key,
    value jsonb not null
); -- goal_status_thresholds, fx_provider, invite_ttl_days ...

```

9.4 Derived views and functions

Object	Type	Purpose
goal_balances	view (security_invoker)	goal_id, balance_minor = Σ goal_amount_minor, contribution_count, last_contribution_date. Never stored (BR-08).
period_summary(p_start, p_end)	SQL function, security invoker	Returns income, expenses, recorded savings for auth.uid() in base currency. Used by Home and Insights.
category_breakdown(p_start, p_end)	SQL function	Expense totals per category for the period.
spending_series(p_start, p_end, p_bucket)	SQL function	Bucketed income/expense series for charts (day/week/month).
accept_invitation(p_token)	SQL function, security definer	Atomic: verify token hash, expiry, invitee identity match, no open couple for invitee; insert member; set couple active; mark invitation accepted.
end_couple()	SQL function, security definer	Set couple ended, left_at for both members, archived_at on couple goals.
sync_goal_completion()	trigger on goal_contributions	Sets/clears savings_goals.completed_at when balance crosses target (BR-11).

Object	Type	Purpose
recalc_base_amounts(p_user)	SQL function (job)	Recomputes base_amount_minor and contributor_base_amount_minor after base-currency change.

9.5 Row-Level Security policy matrix

RLS is enabled on every table. Helper: `is_couple_member(c uuid)` returns true when `auth.uid()` has an open or ended membership in couple `c` (ended members keep read access to history, BR-18).

Table	SELECT	INSERT	UPDATE	DELETE
profiles	own row; partner's <i>name</i> only via view <code>partner_profile</code>	via auth trigger	own row	not permitted from client
transactions	<code>user_id = auth.uid()</code>	<code>user_id = auth.uid()</code>	own rows	own rows
categories	<code>user_id</code> is null OR own	own (<code>is_default = false</code>)	own custom	denied (archive instead)
savings_goals	own individual; couple goals where <code>is_couple_member</code>	own individual; couple goal only if couple active	individual: owner; couple: creator, and couple active	same as update
goal_contributions	goal visible to user	goal visible, couple active (if couple goal), <code>user_id = auth.uid()</code>	own rows, goal not archived	own rows, goal not archived
couples / couple_members	own couple	via functions only	via functions only	via <code>end_couple()</code> only
couple_invitations	<code>inviter = auth.uid()</code>	inviter, via service	inviter (cancel)	denied
exchange_rates, currencies	all authenticated	service role	service role	service role

Rule

AC12 is enforced here: there is no policy anywhere that lets a partner read another user's transactions, individual goals or profile financial fields. pgTAP tests in Section 23 assert this for every table.

10. API Architecture

10.1 Conventions

Aspect	Rule
Base path	/api/v1 — versioned; breaking changes create /api/v2.
Format	JSON, UTF-8. Field names snake_case. Dates ISO 8601 (date: YYYY-MM-DD; timestamps UTC with Z).
Money	Objects: {"amount_minor": 1250, "currency": "USD", "formatted": "\$12.50"}. Clients never parse "formatted".
Auth	Web: Supabase session cookie (HTTP-only, Secure, SameSite=Lax). Native clients: Authorization: Bearer <access_token>. Both resolved by one getSession() guard.
CSRF	Mutating routes require same-origin (Origin header check) for cookie sessions.
Idempotency	POST creating transactions/contributions accepts Idempotency-Key (UUID). Replays within 48 h return the original response.
Pagination	Cursor-based: ?limit=50&cursor=<opaque>. Response: {data:[], next_cursor}.
Rate limits	Auth routes: 5 attempts / 15 min / identifier + 20 / 15 min / IP. Other writes: 60 / min / user. 429 with Retry-After.
Errors	Consistent envelope (10.2). No stack traces or SQL in responses.
Authorization	Enforced twice: application guard + RLS. A missing or foreign resource returns 404 (not 403) to avoid existence leaks.

10.2 Error envelope

```
HTTP 422
{
  "error": {
    "code": "VALIDATION_FAILED",           // stable, machine-readable
    "message": "Some fields need attention.",
    "fields": { "amount": "Enter an amount greater than 0." },
    "request_id": "req_01J9..."
  }
}
Codes: VALIDATION_FAILED 422 | UNAUTHENTICATED 401 | NOT_FOUND 404 | CONFLICT 409
       COUPLE_REQUIRED 409 | GOAL_ARCHIVED 409 | RATE_LIMITED 429 | FX_UNAVAILABLE 503 | INTERNAL 500
```

10.3 Endpoint catalogue — PRD §10

Method & path	Purpose	Notes
POST /auth/register	Create account	{name, email phone, password}. 409 CONFLICT on duplicate (generic message).
POST /auth/login	Start session	Sets cookies (web) or returns tokens (native).
POST /auth/logout	End session	Current device.
POST /auth/refresh	Refresh tokens	Native clients; web refresh handled by middleware.
GET /transactions	List own transactions	Filters: type, category_id, from, to, q, currency; cursor pagination.
POST /transactions	Create	{type, amount_minor, currency, category_id, transaction_date, note?, id?}. Server converts.
GET /transactions/:id	Detail	
PATCH /transactions/:id	Edit	Re-converts if amount, currency or date change.

Method & path	Purpose	Notes
DELETE /transactions/:id	Soft delete	Returns 200 {undo_until}. Purged after 30 days.
GET /insights/daily weekly monthly	Period insights	?date=YYYY-MM-DD anchors the period (default today). Includes previous-period block.
GET /goals	List	?scope=mine ours all&include=completed
POST /goals	Create	{type, name, target_amount_minor, currency, target_date, icon?}. 409 COUPLE_REQUIRED.
GET /goals/:id	Detail + computed metrics	balance, remaining, progress, required pace, current pace, projection, status, contributor breakdown.
PATCH /goals/:id	Edit name, target, date, icon	Currency immutable (BR-15).
DELETE /goals/:id	Delete goal and its contributions	Couple goal: creator only.
POST /goals/:id/contributions	Add contribution	Idempotent. 409 GOAL_ARCHIVED after couple ended.
GET /goals/:id/contributions	History	Contributor name + amounts in original and goal currency.
GET /couple	Couple state	{status, partner:{name}, invitation?, shared_goal_count}
POST /couple/invite	Invite	{invitee: email phone}. Creates pending couple if none.
POST /couple/invitations/:id/accept	Accept	Body {token}. Calls accept_invitation().
DELETE /couple	End couple	BR-18.

10.4 Implementation endpoints (required by screens listed in the PRD)

Method & path	Required by
GET / PATCH /me	Profile screen; "set default currency" step of the auth flow; timezone.
POST /auth/forgot-password, POST /auth/reset-password, POST /auth/verify	Forgot Password screen (design spec §3, §7).
GET / POST /categories, PATCH /categories/:id	Profile → Categories (PRD §9) and category pickers.
GET /activity	Activity screen "Savings Contributions" filter: merged feed of own transactions + own contributions.
GET /home/summary?period=today week month	Home dashboard in one round trip (performance target).
GET /currencies, GET /exchange-rates?date=	Currency pickers; client-side conversion preview and offline estimates.
PATCH /goals/:id/contributions/:cid, DELETE .../:cid	FR-15 (correcting mistaken contributions).
POST /couple/invitations/:id/cancel resend decline	Couple screen invitation states (design spec §19).
GET /invitations/by-token/:token	Public invitation landing page (inviter first name only).

10.5 Example: goal detail response

```
GET /api/v1/goals/6b1e...
{
  "id": "6b1e...", "type": "individual", "name": "New Laptop", "icon": "laptop",
  "currency": "USD",
  "target": { "amount_minor": 120000, "currency": "USD", "formatted": "$1,200.00" },
  "balance": { "amount_minor": 60000, "currency": "USD", "formatted": "$600.00" },
  "remaining": { "amount_minor": 60000, "currency": "USD", "formatted": "$600.00" },
  "progress_pct": 50.0,
  "target_date": "2026-12-31", "days_remaining": 105,
  "required_pace": { "daily": 571, "weekly": 4000, "monthly": 17393, "overdue": false },
  "current_pace_daily": 769,
  "projected_completion_date": "2026-12-05",
  "status": "on_track",
  "contributors": null,
  "completed_at": null, "archived_at": null
}
```

11. Multi-Currency Architecture

Approved scope exception. Design goal: users enter money in whatever currency they actually received or spent, and every total they see is in one currency they chose, with numbers that never silently change.

11.1 Rate source and sync

Item	Specification
Primary provider	ExchangeRate-API Open Access endpoint (daily, no key, attribution required). Attribution link shown in Settings → Currency and in the site footer.
Fallback provider	fxfeed.io free tier (API key). Used automatically if the primary fails or omits a currency.
Pre-build check	Before launch, confirm every currency in the seed list (including LRD) is returned by the chosen provider. Any currency without a rate is set <code>is_active = false</code> .
Schedule	<code>pg_cron</code> at 01:00 UTC daily invokes Edge Function <code>fx-sync</code> : fetch USD-based rates → upsert <code>exchange_rates</code> for today. Retries at 01:30 and 03:00 on failure.
Provider abstraction	<code>interface FxProvider { latest(): Promise<Record<Code, Decimal>> } — swapping providers is a config change in <code>app_config.fx_provider</code>.</code>
Staleness	If no rate stored in 72 h, admin alert (Sentry). Users see "Rates last updated ..." in the currency preview.

11.2 Conversion rules

Event	What is converted	Rate used
Create/edit transaction	amount → user base currency (<code>base_amount_minor</code>)	Rate for <code>transaction_date</code> (F-24)
Create/edit contribution	amount → goal currency (<code>goal_amount_minor</code>) and → contributor's base (<code>contributor_base_amount_minor</code>)	Rate for <code>contribution_date</code>
Display partner contributions	Nothing: shown in goal currency and original currency	Stored values
Base currency change	All of the user's <code>base_amount_minor</code> and <code>contributor_base_amount_minor</code>	Each record re-converted from its original amount at the rate of its own date (not today's)
Offline entry	Client shows estimate from cached rates; server performs authoritative conversion on sync	Rate for record date

11.3 Base currency change

1. User selects a new currency; dialog explains: "All your totals will be shown in EUR. Past entries are converted at the rate from their own dates. Your original amounts are kept."
2. PATCH /me sets `base_currency` and enqueues `recalc_base_amounts(user)` (runs in one transaction; target < 5 s for 10,000 records).
3. While running, API responses carry `recalculating: true`; UI shows a banner and keeps prior values.
4. Goal currencies are not changed (BR-15).

11.4 Display rules

- Amounts are formatted with `Intl.NumberFormat` using the user's locale and the currency code; symbols alone are ambiguous (\$ is USD and LRD), so non-base amounts always show the ISO code: **L\$ 5,000.00 LRD / \$26.40**.
- Rows entered in a non-base currency show the original amount on the first line and "≈ base amount" underneath in secondary text.

- fx_estimated records show an info icon: "Converted using the closest available rate."
- Totals are always in base currency and labelled with the code once per card header (e.g. "This month · USD").

PART C — EXPERIENCE

12. Web Information Architecture

12.1 Sitemap and routes

Route	Screen	Access	Notes
/	Welcome / landing	Public	Redirects to /home when a valid session exists.
/onboarding	Onboarding (3 pages)	Public	Shown once per device (flag in localStorage — not sensitive). Skippable.
/register · /login	Register · Login	Guest only	?next= return URL (same-origin only).
/forgot-password · /reset-password · /verify	Password reset · verification	Guest	Tokens from email/SMS.
/setup/currency	Base currency + timezone	Auth, not onboarded	Required once before /home.
/home?period=today week month	Home dashboard	Auth	Default period: month.
/add/income · /add/expense	Add forms	Auth	Rendered as dialog (desktop) / full sheet (mobile) via intercepted route; direct URL renders full page.
/activity?type=&category=&from=&to=&q=	Activity	Auth	Filters live in the URL (shareable, back-button safe).
/activity/[id] · /activity/[id]/edit	Transaction details · edit	Auth	Desktop: side panel; mobile: pushed page.
/insights?period=daily weekly monthly&date=	Insights	Auth	
/goals?tab=mine ours	Goals	Auth	
/goals/new?type=individual couple	Create goal	Auth	
/goals/[id] · /goals/[id]/edit · /goals/[id]/contribute	Goal details · edit · add contribution	Auth / couple member	
/couple	Couple	Auth	
/invite/[token]	Invitation landing	Public	Shows inviter first name; routes to register/login, then accept.
/profile · /profile/categories · /profile/currency · /profile/notifications · /profile/security	Profile & settings	Auth	
/offline	Offline fallback	Public	Served by service worker when an uncached route is requested offline.

12.2 Navigation model

Viewport	Primary navigation	Global Add	Detail views
< 768 px (mobile)	Bottom tab bar: Home, Activity, Goals, Insights, Profile	Floating action button bottom-right, 16 px above tab bar → bottom sheet with Income / Expense / Savings contribution	Pushed pages with back button; forms as full-height sheets
768–1023 px (tablet)	Left icon rail (72 px) with tooltips + labels on focus	"+" button at top of rail	Pushed pages; forms as centred dialogs (max 480 px)

Viewport	Primary navigation	Global Add	Detail views
≥ 1024 px (desktop)	Left sidebar (240 px) with labels, user card at bottom	Primary "Add" button at top of sidebar; keyboard shortcut N	Activity uses list + details split view; forms as dialogs

- The Add sheet offers **Add income, Add expense, Add savings contribution** (the last opens a goal picker, then the contribution form). Reachable from every main tab (design spec §9).
- Browser back/forward MUST work for every navigation, including opening and closing dialogs (intercepted routes handle this).
- Keyboard shortcuts (desktop): N = new transaction, / = focus search on Activity, G then H/A/G/I/P = go to section. Listed in a "?" help dialog; all shortcuts can be disabled in Profile.

13. Screen Specifications

Every screen below also inherits the global loading, empty, error and offline rules in Section 19 and the accessibility rules in Section 20. Wireframes for the key screens follow in Section 15.

SCR-01 Splash / session restore (any)

Purpose	Restore session before rendering app routes.
Content (top → bottom)	Brand mark centred; thin progress indicator after 400 ms.
Interactions & states	Valid session → /home (or setup if not onboarded). No session → Welcome. Network failure → "Can't reach SpendTogether" + Retry; never a dead end.

SCR-02 Welcome /

Purpose	Communicate the value proposition and route to account creation.
Content (top → bottom)	Brand, tagline, illustration, three-line promise, Create account (primary), Log in (secondary). Desktop: split layout, illustration right.
Interactions & states	Create account → /register; Log in → /login.

SCR-03 Onboarding /onboarding

Purpose	Explain the three core ideas in ≤ 3 pages.
Content (top → bottom)	Pager: 1) Track income and expenses 2) Understand daily/weekly/monthly patterns 3) Save alone or together. Illustration, headline, one sentence, dots, Skip, Next / Get started.
Interactions & states	Swipe or arrow keys between pages. Skip → /register.

SCR-04 Register /register

Purpose	Create an account.
Content (top → bottom)	Back, title, Name, Email or phone (auto-detects format), Password (show/hide, strength hint: min 10 chars), Confirm password, Create account, link to Log in.
Interactions & states	Validation after blur. Submit disabled + inline spinner while pending. Duplicate → generic conflict message with Log in link.

SCR-05 Login /login

Purpose	Authenticate.
Content (top → bottom)	Identifier, Password (show/hide), Forgot password link, Log in, link to Register.
Interactions & states	Auth error shown above the button; does not reveal which field was wrong. After 5 failures: "Too many attempts. Try again in 15 minutes."

SCR-06 Forgot / reset password /forgot-password, /reset-password

Purpose	Recover access.
Content (top → bottom)	Step 1: identifier → "If an account exists, we've sent instructions." Step 2: new password + confirm.
Interactions & states	Always the same confirmation message (no account enumeration). On success all other sessions are revoked.

SCR-07 Currency setup /setup/currency

Purpose	Choose base currency and confirm timezone.
Content (top → bottom)	Explanation, searchable currency list (code, name), pre-selected from locale, detected timezone with Change link, Continue.
Interactions & states	Continue → PATCH /me → /home (empty state).

SCR-08 Home dashboard /home

Purpose	Show the current period's financial position without requiring a chart.
Content (top → bottom)	Greeting + date + base currency; period selector (Today / This week / This month); hero card "Remaining this [period]" with Income, Expenses, Saved, Savings rate; net cash flow line; spending preview (top 5 categories with amount, %, bar); active goals preview (max 3); recent activity (desktop, 5 rows); quick add.
Interactions & states	Tap category → /activity filtered by category and period. Tap goal → goal details. Tap metric → /insights for that period.

SCR-09 Add sheet (overlay)

Purpose	Choose what to add.
Content (top → bottom)	Three rows, each icon + title + one-line explanation: Income ("Money you received"), Expense ("Money you spent"), Savings contribution ("Money you put toward a goal").
Interactions & states	Selecting opens the matching form. Esc / swipe down / backdrop click closes.

SCR-10 Add income /add/income

Purpose	Record income fast.
Content (top → bottom)	Header, Amount (dominant, numeric keypad, autofocus), currency chip (default base), conversion preview if non-base, Category (bottom sheet picker, searchable), Date (default today), Note (optional, 280 chars), Save income.
Interactions & states	Save disabled until amount > 0 and category chosen. Success: close, toast "Income added", optimistic totals. Failure: values kept, error + Retry.

SCR-11 Add expense /add/expense

Purpose	Record an expense fast.
Content (top → bottom)	Identical structure to SCR-10 for muscle memory; recent categories (last 5 used) at the top of the picker.
Interactions & states	Same as SCR-10. No celebratory motion (design spec §22).

SCR-12 Activity /activity

Purpose	Find, open, edit and delete records.
Content (top → bottom)	Title, search, filter chips (All, Income, Expense, Savings contributions, Category, Date range), date-grouped list. Row: category icon, title (category or goal name), note, signed amount with +/- and type icon; non-base rows show "≈ base".
Interactions & states	Row → details (split panel on desktop). Infinite scroll (50 per page). Clear filters in empty result.

SCR-13 Transaction details / edit / delete /activity/[id]

Purpose	Inspect and change one record.
Content (top → bottom)	Type and amount (original + converted with rate and date), category, date, note, Edit, Delete. Edit reuses the Add form.
Interactions & states	Delete → confirmation dialog stating summaries will be recalculated → toast with Undo (5 s).

SCR-14 Insights /insights

Purpose	Explain patterns by day, week and month.
Content (top → bottom)	Period control (Daily / Weekly / Monthly) + date stepper (< September 2026 >); 4 metric cards with previous-period change; spending trend; income vs expenses; category breakdown; savings rate; average daily spending.
Interactions & states	Changing period refreshes every card consistently. Category → filtered Activity. Each chart has "View as table".

SCR-15 Goals /goals

Purpose	Show progress on all goals.
Content (top → bottom)	Segment My goals / Our goals; goal cards (icon, name, saved of target, progress bar, %, target date, status chip with icon + text); Completed goals collapsed section; Create goal.
Interactions & states	Our goals with no partner → empty state + Connect partner. Partner but no goals → Create shared goal.

SCR-16 Create / edit goal /goals/new

Purpose	Define a target.
Content (top → bottom)	Goal type (Individual default / Couple — disabled with explanation if no active couple), name, icon picker, target amount + currency, target date, Create goal.
Interactions & states	Target > 0, date ≥ today. Success → goal details with empty contribution state (no forced contribution).

SCR-17 Goal details /goals/[id]

Purpose	Make progress and required effort obvious.
Content (top → bottom)	Hero: name, saved, target, progress bar, %; remaining; required pace (per day/week/month); status card with plain-language sentence; projected completion; contributor breakdown (couple); contribution history; persistent Add contribution button (sticky on mobile).
Interactions & states	Zero contributions: \$0 saved, remaining, required pace, Add contribution. Archived (couple ended): banner, no add button.

SCR-18 Add contribution /goals/[id]/contribute

Purpose	Fund a goal.
Content (top → bottom)	Goal context (name, saved of target), Amount + currency (default goal currency), conversion preview, Date, Note, "After this" preview, Add contribution.
Interactions & states	Target reached → completion state: restrained confetti (≤ 1.2 s, off with reduced motion), status Completed, haptic on supported devices.

SCR-19 Couple /couple

Purpose	Manage the partner connection.
Content (top → bottom)	States: No partner (explanation + Invite partner); Pending (destination, expiry, Resend, Cancel); Connected (partner name, since date, shared goals preview, Create shared goal, End connection).
Interactions & states	Never displays partner income, expenses, balances or individual goals.

SCR-20 Invitation landing /invite/[token]

Purpose	Let the invitee join.
Content (top → bottom)	"Alex invited you to save together on SpendTogether." Privacy explanation. Accept / Decline (if logged in) or Create account / Log in.
Interactions & states	Invalid/expired token → explanation + ask inviter to resend. Invitee already in a couple → cannot accept, with reason.

SCR-21 Profile & settings /profile

Purpose	Manage account and preferences.
Content (top → bottom)	Profile summary; Preferences: base currency, timezone, categories, notifications; Couple settings; Security: change password; Log out.
Interactions & states	Base currency change → impact dialog (Section 11.3). Destructive actions confirmed with consequences.

SCR-22 Categories /profile/categories

Purpose	Manage custom categories.
Content (top → bottom)	Tabs Expense / Income; default categories (locked icon); custom categories with icon, name, Edit, Archive; Add category.
Interactions & states	Archiving hides from pickers; history keeps the category.

14. UI Component Hierarchy

14.1 Application tree

```

<RootLayout>                                html lang, fonts, theme tokens, <Toaster/>
├── <Providers>                               QueryClientProvider · SessionProvider · OfflineProvider
│   ├── (public) <PublicLayout>               Welcome · Onboarding · Invite landing
│   ├── (auth) <AuthLayout>                   centred card, back link
│   └── (app) <AppShell>
│       ├── <SkipLink/>
│       ├── <Sidebar/> | <IconRail/> | <BottomTabBar/>    (by breakpoint)
│       ├── <AddButton/> → <AddSheet/>                global add
│       ├── <OfflineSyncIndicator/>                  header chip
│       ├── <RecalculatingBanner/>                    base-currency change
│       ├── <main id="content"> {page} </main>
│       ├── {@modal slot} <DialogSheet> <TransactionForm/> | <ContributionForm/>
│       └── <ConfirmationDialog/>                    imperative, one instance

```

14.2 Screen compositions

```

HomePage
├── <PageHeader greeting date currency> <PeriodSelector/>
├── <SummaryHeroCard> <SummaryMetric/> *4 <NetCashFlowLine/>
├── <SpendingPreview> <CategoryRow/> *≤5 <ViewAllLink/>
├── <GoalsPreview> <GoalCard compact/> *≤3
└── <RecentActivity> <TransactionRow/> *5 (≥1024 px only)

ActivityPage
├── <PageHeader/> <SearchInput/> <FilterChipGroup/> <DateRangePicker/>
├── <SectionList> <DateSectionHeader/> <TransactionRow/>... <InfiniteLoader/>
└── <DetailPanel> <TransactionDetails/> (≥1024 px split view)

InsightsPage
├── <PeriodControl/> <DateStepper/>
├── <MetricCard/> *4 (value + <DeltaChip/>)
├── <ChartContainer title> <SpendingTrendChart/> <DataTableToggle/>
├── <ChartContainer> <IncomeExpenseChart/>
├── <ChartContainer> <CategoryDonut/> <CategoryLegendList/>
└── <MetricCard savings-rate/> <MetricCard avg-daily/>

GoalDetailsPage
├── <GoalHero> <ProgressBar/> <MoneyText/>
├── <PaceCard/> <StatusCard/> <ProjectionText/>
├── <ContributorBreakdown/> couple only
├── <ContributionList> <ContributionRow/>...
└── <StickyActionBar> <PrimaryButton>Add contribution</PrimaryButton>

```

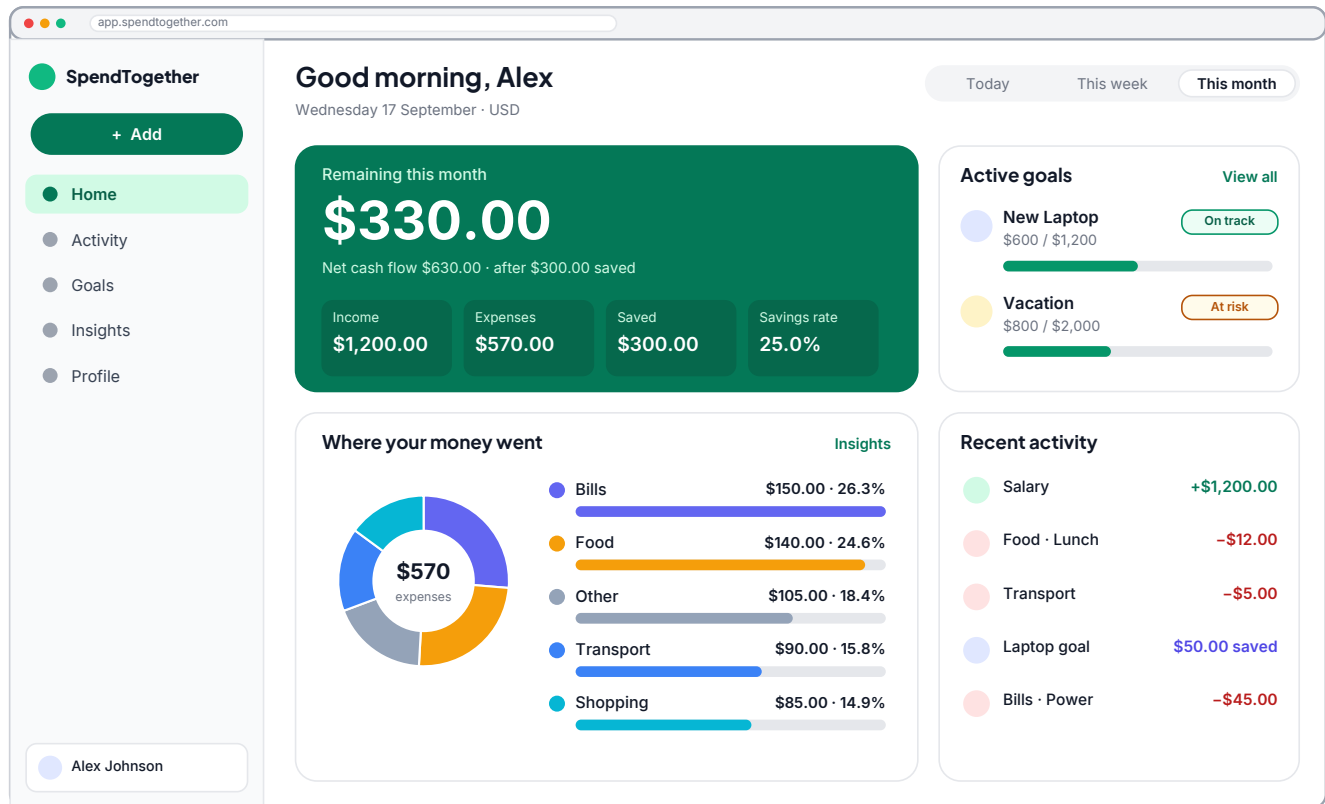
14.3 Shared component catalogue

Component	Key props	Behaviour / rules
Button	variant: primary secondary tertiary destructive ghost; size: sm md lg; loading; iconLeft	Min height 44 px (md). Loading keeps width, shows spinner, sets aria-busy.
IconButton	icon, label (required)	label becomes aria-label and tooltip.
Input / Textarea	label, hint, error, maxLength	Label always visible; error linked via aria-describedby; 16 px text (prevents iOS zoom).
AmountInput	value (minor), currency, onCurrencyChange, previewBase?	Locale-aware decimal entry, currency exponent respected, inputmode="decimal", shows ≈ conversion line.

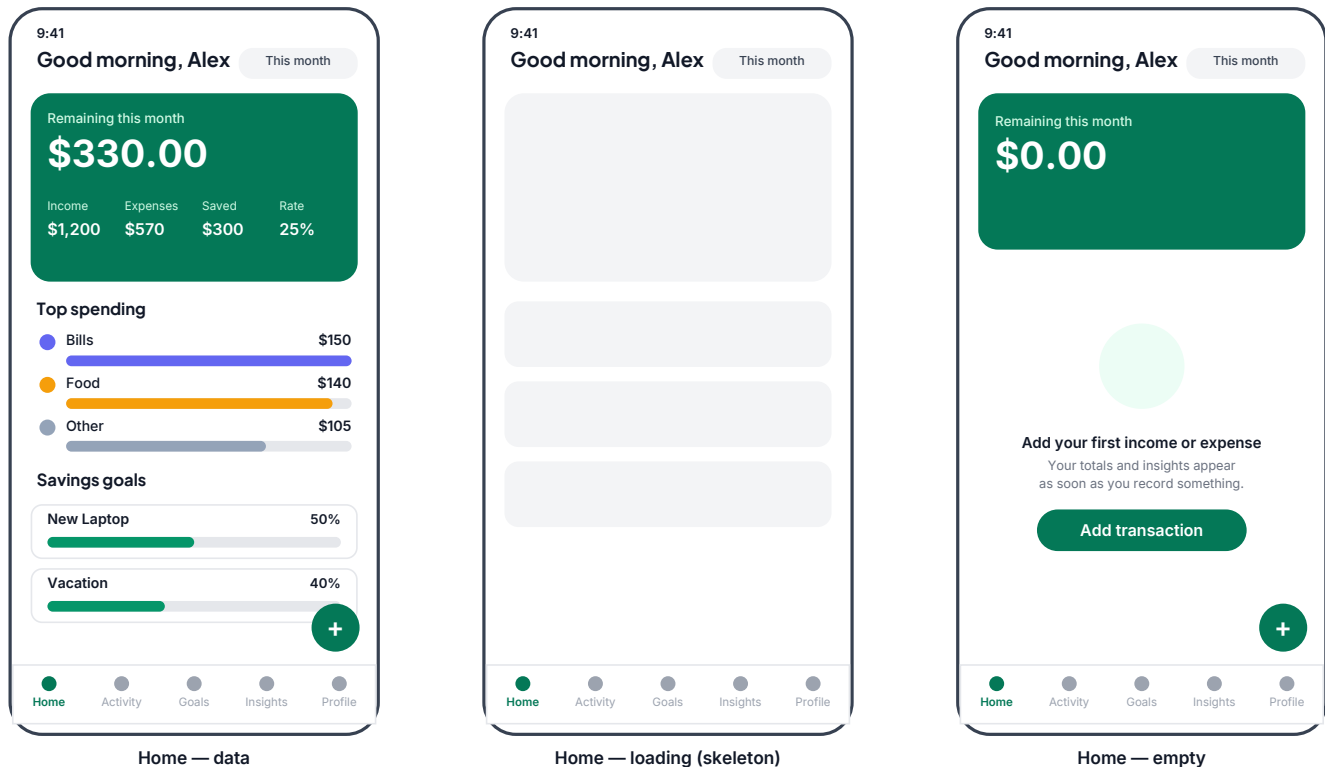
Component	Key props	Behaviour / rules
CurrencyPicker	value, recent[]	Searchable list; recent + base pinned on top.
CategoryPicker	type, value, recent[]	Sheet on mobile, popover on desktop; icon + name; search.
DatePicker	value, max=today	Native date input on mobile; calendar popover on desktop; "Today"/"Yesterday" shortcuts.
SelectRow	label, value, onPress	Settings rows with chevron.
PeriodSelector / SegmentedControl	options, value	role="radiogroup"; arrow-key navigation.
SummaryMetric / MetricCard	label, money pct, delta?	Numbers use tabular figures; delta chip shows arrow icon + text, not colour alone.
TransactionRow	transaction, onPress	Icon, title, note, signed amount with type icon; whole row is one button with full accessible name.
GoalCard	goal, compact?	Name, saved of target, ProgressBar, %, date, StatusChip.
ProgressBar	value 0–100, label	role="progressbar" with aria-valuenow/-text.
StatusChip	status	Icon + text: ✓ On track, ! At risk, ↓ Behind, ★ Completed.
BottomSheet / Dialog	open, title, onClose	Radix Dialog; focus trap; Esc; returns focus to trigger.
ConfirmationDialog	title, consequences, confirmLabel, destructive	Destructive button right; Cancel is default focus.
Toast	message, action?	role="status"; 5 s; action (Undo) keyboard reachable; pauses on hover/focus.
EmptyState	illustration, title, body, action	Answers: what is missing, why it matters, what to do next.
LoadingSkeleton	shape	Matches final layout; shimmer disabled with reduced motion.
ErrorState	message, onRetry	Plain language + Retry; keeps cached content visible when possible.
ChartContainer	title, description, data table	figure + figcaption; "View as table" toggle.
FilterChip	label, selected	aria-pressed.
OfflineSyncIndicator	pendingCount	"Sync pending (2)" chip; tap shows queue details.

15. Wireframe Sketches

Low-fidelity layouts that fix structure, hierarchy and content. They use the corrected reference dataset and the accessible colour shades. Visual polish follows the tokens in Sections 17–18.



W-01 Home dashboard — desktop (≥ 1024 px): sidebar, hero "Remaining this month", goals, category spending, recent activity.



W-02 Home dashboard — mobile (< 768 px): data, skeleton loading, and first-run empty states.

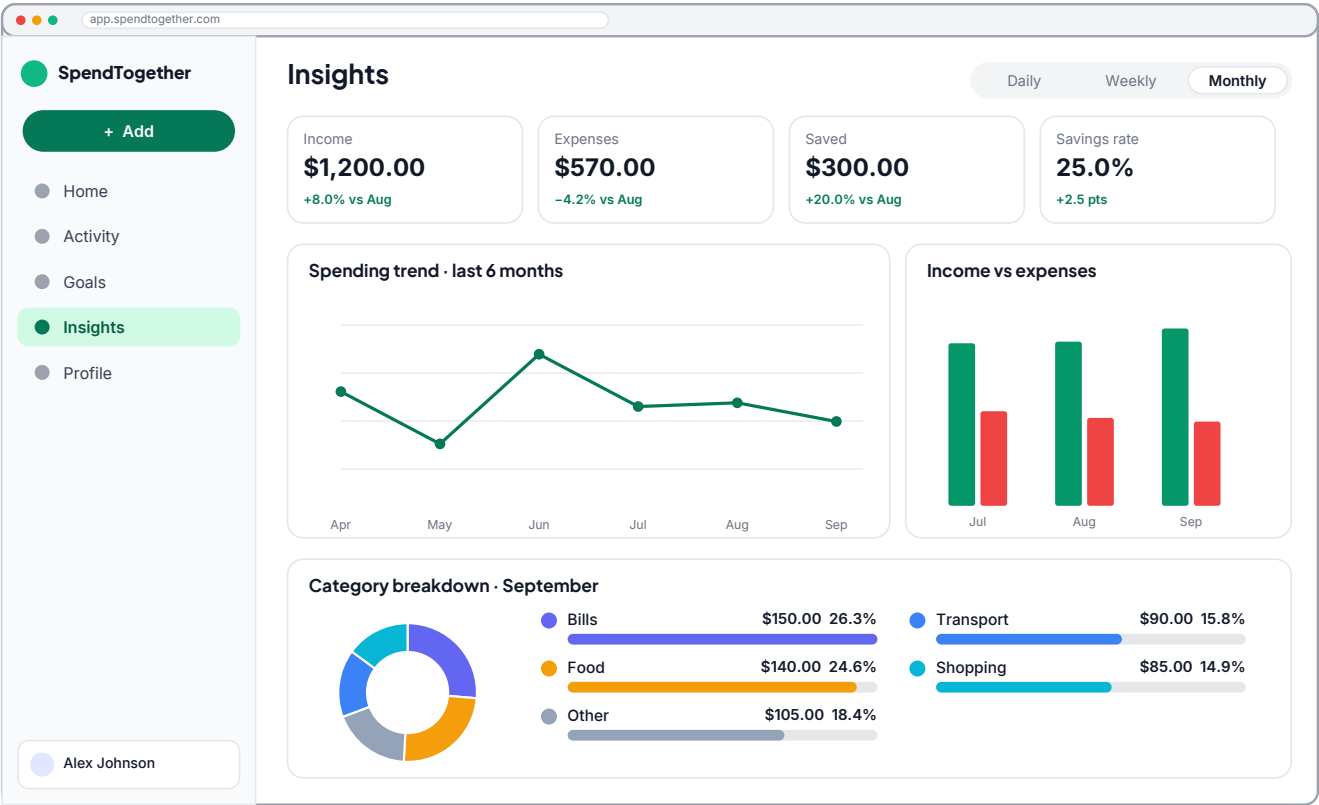
The left sketch shows a desktop dialog titled 'Add expense' with a close button. It contains fields for Amount (12.00), Currency (USD), Category (Food), Date (Today, 17 Sep 2026), and Note (optional) (Lunch). A 'Save expense' button is at the bottom. The right sketch shows a mobile full-height sheet titled 'Add expense' with a close button. It features tabs for Income and Expense, a large display for L\$ 5,000.00, a note about the rate (≈ \$26.40 USD · rate of 17 Sep), a Category field (Transport), a Date field (Today), a Note field (Taxi to work), and a 'Save expense' button.

Mobile: full-height sheet

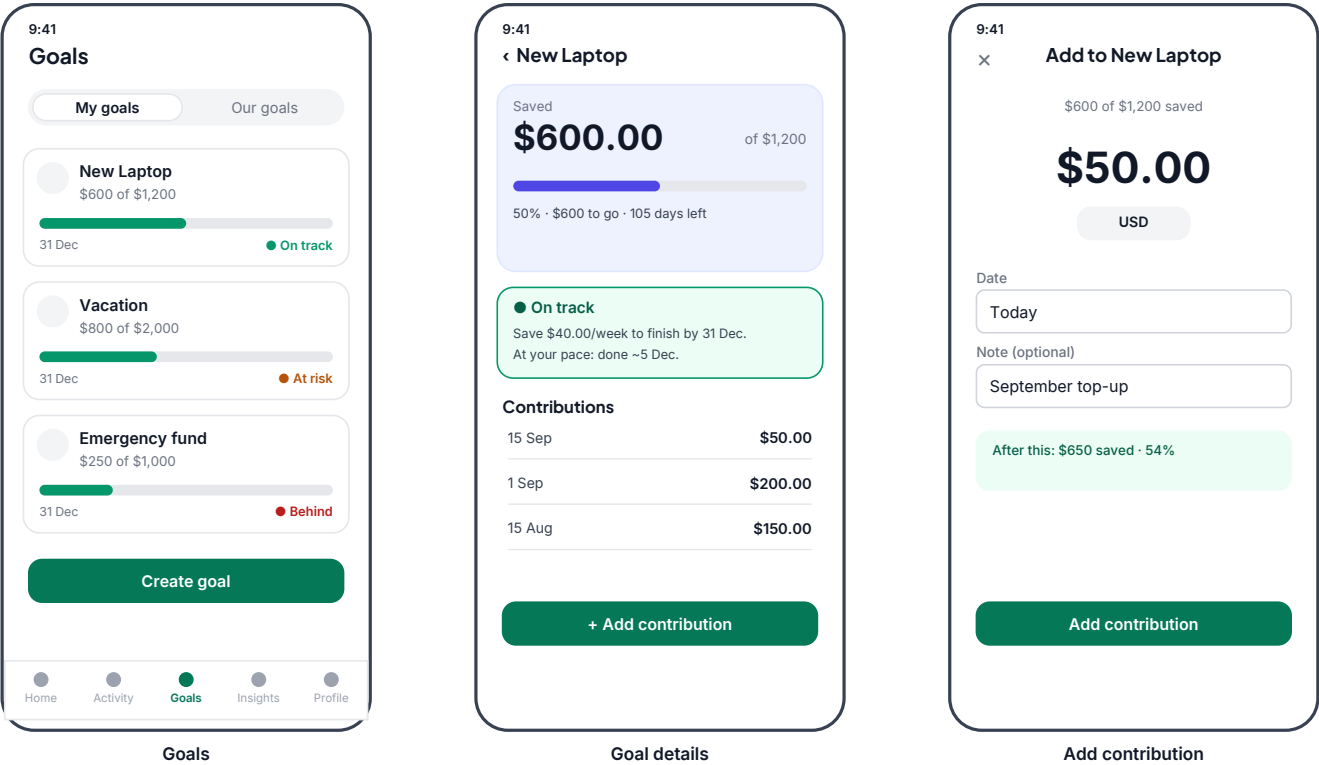
W-03 Add expense — desktop dialog (left) and mobile full-height sheet with a non-base currency (right). Rate shown is illustrative.

The sketch shows a desktop split view for the 'Activity' section. On the left is a sidebar with 'SpendTogether' and a '+ Add' button, and a list of navigation items: Home, Activity (selected), Goals, Insights, and Profile. The main area has a search bar and filter chips for All, Income, Expense, Savings, Category, and Date. It displays a date-grouped list of transactions for Wednesday, 17 September (Food · Lunch: -\$12.00, Transport · Taxi: -L\$ 5,000 LRD ≈ -\$26.40, Laptop goal: \$50.00 saved) and Tuesday, 16 September (Salary: +\$1,200.00, Bills · Electricity: -\$45.00, Shopping: -\$35.00). On the right is a 'Transaction details' panel for the selected 'Expense · Transport' transaction, showing L\$ 5,000.00 LRD, the rate (≈ \$26.40 USD (rate 1 USD = 189.39 LRD, 17 Sep)), and fields for Category (Transport), Date (17 Sep 2026), and Note (Taxi to work). 'Edit' and 'Delete' buttons are at the bottom.

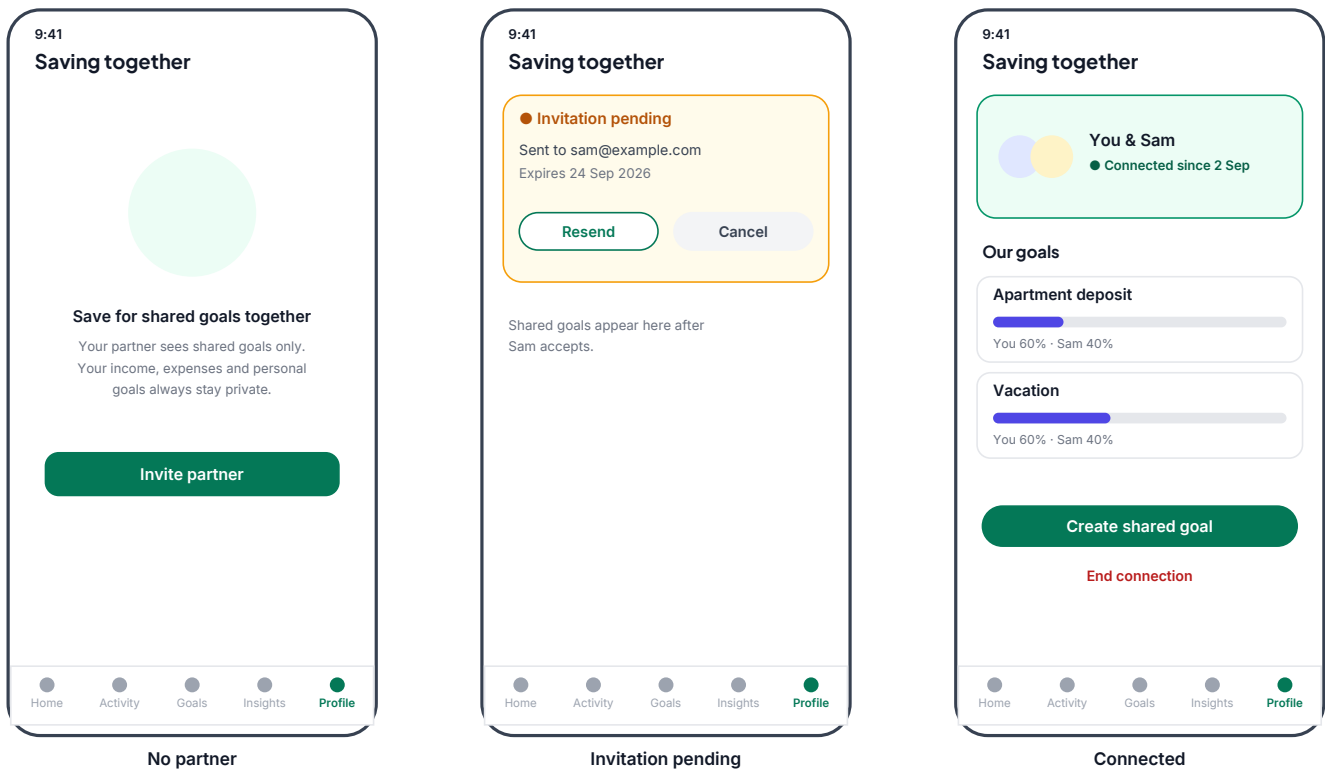
W-04 Activity — desktop split view: search, filter chips, date-grouped list, details panel with conversion info.



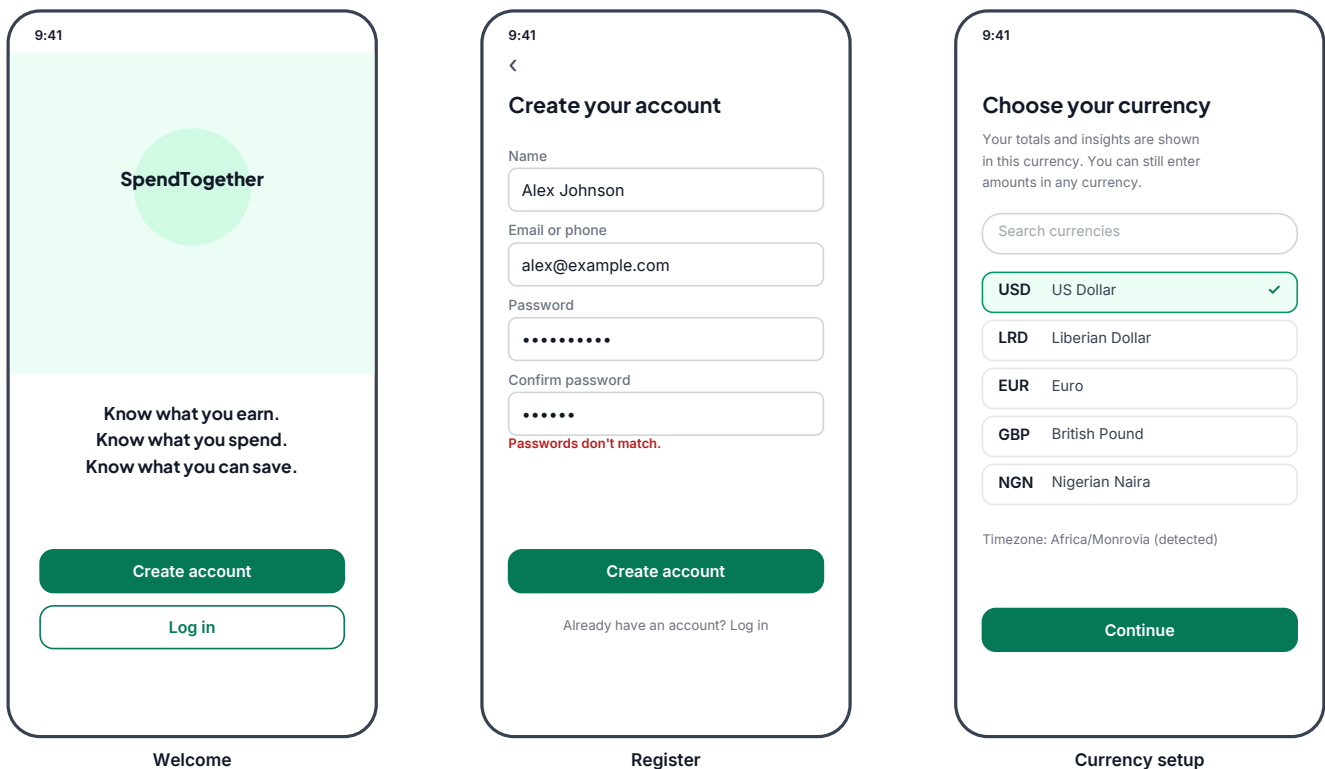
W-05 Insights — monthly view: metric cards with previous-period change, trend, income vs expenses, category breakdown.



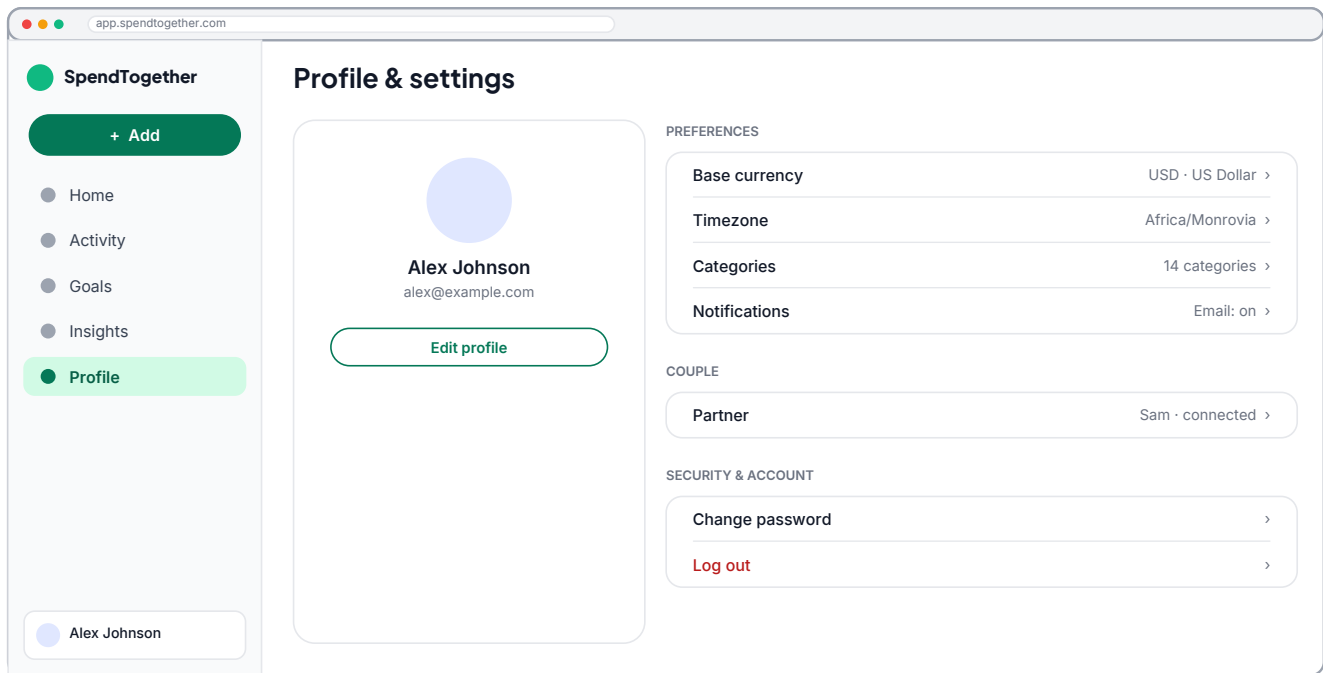
W-06 Goals list, goal details and add contribution — mobile.



W-07 Couple screen — the three required states. No private data appears in any state.



W-08 Welcome, Register (inline validation), and base-currency setup — mobile.



W-09 Profile & settings — desktop.

16. Charts & Analytics Specifications

Charts support understanding; they never replace numbers. The Home screen communicates the period position without any chart (design acceptance #3). All charts are built with Recharts inside a shared ChartContainer.

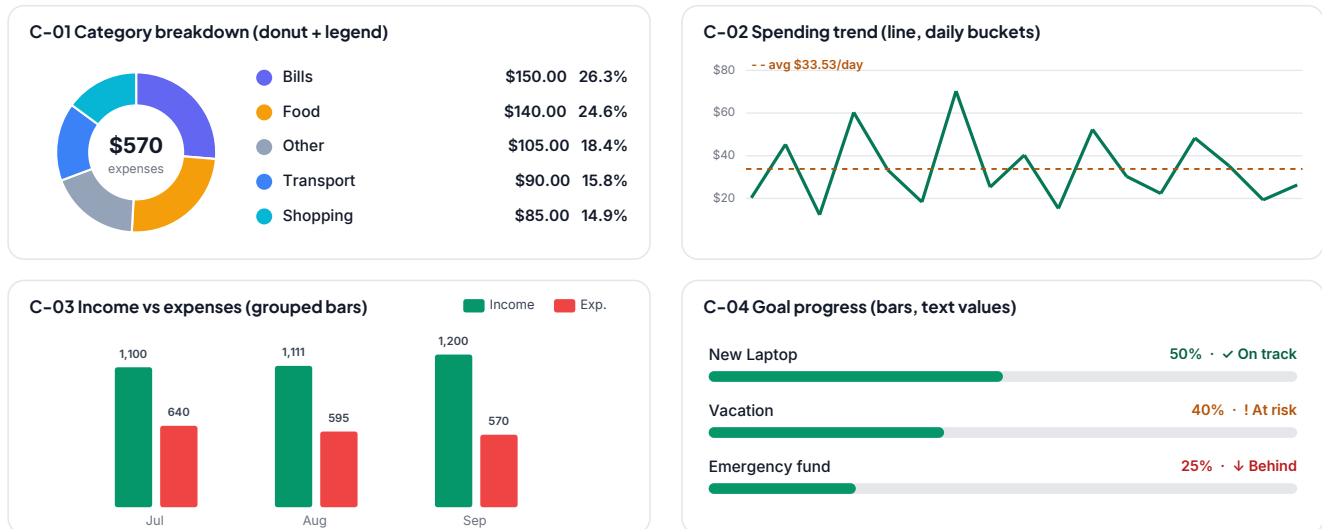


Chart specimens using the corrected reference dataset.

16.1 Chart catalogue

ID	Chart	Where	Data	Encoding
C-01	Category breakdown	Home (list only), Insights	F-08, F-09 for the period; top 6 categories + "Other" bucket	Donut (inner radius 60%) + legend list with amount and %; total in centre. Sorted by amount desc.
C-02	Spending trend	Insights	Expense totals per bucket: Daily view → last 14 days; Weekly → last 8 weeks; Monthly → last 6 months	Line, 2 px, dots on hover/focus; dashed average line (F-07); y-axis starts at 0.
C-03	Income vs expenses	Insights	F-01, F-02 per bucket (same buckets as C-02, last 3 shown on mobile, 6 on desktop)	Grouped vertical bars; income left, expenses right; value labels above bars ≥ 768 px.
C-04	Goal progress	Goals, Goal details, Home	F-11...F-13	Horizontal bar, 8 px, rounded; % and status text beside.
C-05	Savings rate	Home, Insights	F-06 current + previous	Metric card with delta chip; "N/A" with explanation when income = 0.
C-06	Previous-period comparison	Insights	F-10 for income, expenses, saved, savings rate	Delta chips: arrow icon + signed % + "vs Aug"; "New" when previous = 0.
C-07	Contributor breakdown	Couple goal details	F-21	Stacked single bar (two segments, secondary + accent) + text "You \$480 (60%) · Sam \$320 (40%)".

16.2 Rules for every chart

- Every chart has a visible title, a one-line description, and a **View as table** toggle rendering an HTML <table> with the same data.
- Values are in base currency with the code in the subtitle ("USD"). Tooltips show formatted money and date range.
- Category colours come from the categorical tokens (Section 18) and are always paired with a text label; hue alone never carries meaning.
- Minimum data: with fewer than 2 buckets containing data, C-02/C-03 show the available totals and the message "Your trends appear as you record more activity" instead of an empty axis.
- No 3D, no dual axes, no pie charts with > 7 slices, no animated counters. Entry animation ≤ 300 ms, disabled with reduced motion.
- Keyboard: data points are focusable in order; focus shows the tooltip. Screen readers get the table, not SVG internals (SVG has aria-hidden="true" when the table is present).

16.3 Insights API contract (per period)

```
GET /api/v1/insights/monthly?date=2026-09-17
{ "period": { "type": "monthly", "start": "2026-09-01", "end": "2026-09-30", "days_elapsed": 17 },
  "currency": "USD",
  "totals": { "income": 120000, "expenses": 57000, "saved": 30000, "net": 63000, "remaining": 33000,
    "savings_rate_pct": 25.0, "avg_daily_spending": 3353 },
  "previous": { "income": 111111, "expenses": 59500, "saved": 25000, "savings_rate_pct": 22.5 },
  "change_pct": { "income": 8.0, "expenses": -4.2, "saved": 20.0, "savings_rate_pts": 2.5 },
  "categories": [ { "id": "...", "name": "Bills", "amount": 15000, "pct": 26.3 }, ... ],
  "series": { "bucket": "month", "points": [ { "start": "2026-04-01", "income": ..., "expenses": ... }, ... ] }
```

16.4 Product analytics events

Instrument the PRD success metrics (§13) with a privacy-respecting analytics tool. Events never contain amounts, notes, category names or partner identity.

Event	Properties	Metric
account_created	method (email/phone)	Activation
base_currency_set	currency	Activation
transaction_created	type, is_foreign_currency, offline_queued	Activation (first), engagement / week
goal_created	type	Activation
contribution_added	goal_type	Engagement / month
partner_invited · invitation_accepted	channel	Partner connection
insights_viewed	period	Insight views
session_start (daily)	—	Retention D7 / D30

PART D — VISUAL DESIGN

17. Colour Palette & Typography

Derived from the SpendTogether boards: calm teal-green primary, indigo secondary, amber accent, neutral grey foundation. Personality: calm, modern, trustworthy, optimistic, non-judgmental.

17.1 Brand and semantic palette

Token	Hex	On white	Allowed use
 primary-500	#10B981	2.54:1 · decorative	Brand fills, illustrations, large icons, progress track accents. Never text.
 primary-600	#059669	3.77:1 · AA large / UI	Progress fills, chart income series, focus-visible outlines (3:1 non-text).
 primary-700	#047857	5.48:1 · AA text	Primary button background (white text 5.48:1), links, income amounts.
 primary-800	#065F46	7.68:1 · AA text	Text on primary-50 tints, success chip text.
 primary-50	#ECFDF5	1.05:1 · decorative	Tinted backgrounds (selected rows, success chips).
 secondary-500	#6366F1	4.47:1 · AA large / UI	Couple/shared accents, savings contribution icons, large text only.
 secondary-600	#4F46E5	6.29:1 · AA text	Savings amounts text, couple chart series.
 accent-500	#F59E0B	2.15:1 · decorative	Highlights, "At risk" chip fill with dark text (8.26:1). Never text on white.
 accent-700	#B45309	5.02:1 · AA text	Warning text, "At risk" text.
 success-500	#22C55E	2.28:1 · decorative	Decorative success only (confetti, illustration).
 error-500	#EF4444	3.76:1 · AA large / UI	Expense chart series, icons.
 error-600	#DC2626	4.83:1 · AA text	Destructive button background (white 4.83:1).
 error-700	#B91C1C	6.47:1 · AA text	Expense amounts text, error messages.
 neutral-900	#111827	17.74:1 · AA text	Headings, primary text, dark header band.
 neutral-700	#374151	10.31:1 · AA text	Body text.
 neutral-600	#6B7280	4.83:1 · AA text	Secondary text, captions (4.83:1).
 neutral-400	#9CA3AF	2.54:1 · decorative	Placeholder icons, disabled — not for meaningful text.
 neutral-300	#D1D5DB	1.47:1 · decorative	Input borders.
 neutral-200	#E5E7EB	1.24:1 · decorative	Dividers, card borders, progress tracks.
 neutral-100	#F3F4F6	1.10:1 · decorative	Skeletons, segmented control track.
 neutral-50	#F9FAFB	1.05:1 · decorative	App background.

Correction to source material

The boards place white text on #10B981 and use #22C55E / #F59E0B as text. Those combinations measure 2.54:1, 2.28:1 and 2.15:1 against white and fail WCAG AA. The brand hues stay; text and filled controls use the -700 shades listed above.

17.2 Financial meaning

Meaning	Text colour	Icon	Sign / label
Income	primary-700	arrow-down-left in circle	+ prefix, "Income"
Expense	error-700	arrow-up-right in circle	– prefix (U+2212), "Expense"
Savings contribution	secondary-600	piggy-bank	"saved" suffix, "Savings"
On track / Completed	primary-800 on primary-50	check / star	text label
At risk	accent-700 on accent-50	alert-triangle	text label
Behind / Overdue / Overspent	error-700 on error-50	arrow-down / clock	text label

17.3 Chart categorical colours

Token	Hex	Default category
 cat-food	#F59E0B	Food
 cat-bills	#6366F1	Bills
 cat-transport	#3B82F6	Transport
 cat-shopping	#06B6D4	Shopping
 cat-health	#EC4899	Health
 cat-education	#8B5CF6	Education
 cat-entertainment	#F97316	Entertainment
 cat-family	#14B8A6	Family
 cat-other	#94A3B8	Other

17.4 Typography

Plus Jakarta Sans (600, 700, 800) for headings and brand. **Inter** (400, 500, 600, 700) for body, UI and every number, with font-feature-settings: "tnum" 1, "cv11" 1 so digits align in lists and totals. Both are served with next/font (self-hosted, no layout shift, display: swap). Fallback stack: system-ui, -apple-system, "Segoe UI", Roboto, sans-serif.

Token	Specimen (scaled)	Family / weight	Size / line (px)	Use
display	Aa \$1,200	Jakarta 800	40 / 48 · 32 / 40 mobile	Hero amount label (numbers use num-display)
num-display	Aa \$1,200	Inter 700 tnum	40 / 44 · 32 / 36 mobile	Hero money value
h1	Aa \$1,200	Jakarta 700	30 / 38	Page titles
h2	Aa \$1,200	Jakarta 700	22 / 30	Section titles, card headings
h3	Aa \$1,200	Jakarta 600	18 / 26	Sub-sections
body-lg	Aa \$1,200	Inter 400	16 / 24	Default body, inputs
body-sm	Aa \$1,200	Inter 400	14 / 20	Secondary text, rows
label	Aa \$1,200	Inter 600	14 / 20	Buttons, labels, metric values
caption	Aa \$1,200	Inter 500	12 / 16	Captions, chart axes (minimum size)

- Text sizes are defined in rem; layout MUST survive 200% browser zoom and OS text scaling without clipping or horizontal scrolling.
- Line length for prose ≤ 72 characters. Money is never truncated with an ellipsis; long values wrap or scale down one step.
- Sentence case everywhere; no all-caps except small overline labels (letter-spacing 0.06em).

18. Design Tokens

Tokens are the only source of visual values. Components reference tokens; screens never hard-code colours, sizes or durations. Defined once as CSS custom properties and exposed to Tailwind.

Group	Tokens
Spacing (4-pt)	space-0 0 · 0.5 2 · 1 4 · 2 8 · 3 12 · 4 16 · 5 20 · 6 24 · 8 32 · 10 40 · 12 48 · 16 64 (px)
Radius	radius-sm 6 · md 10 · lg 14 · xl 20 · full 9999
Elevation	elev-0 none · elev-1 0 1px 2px rgb(17 24 39/.06) · elev-2 0 4px 12px rgb(17 24 39/.08) · elev-3 0 12px 32px rgb(17 24 39/.16) (dialogs, sheets)
Motion duration	dur-fast 120 ms (hover, press) · dur-base 200 ms (sheets, dialogs) · dur-slow 320 ms (progress, page) · dur-celebrate 1200 ms (goal completed)
Motion easing	ease-standard cubic-bezier(.2,0,0,1) · ease-exit cubic-bezier(.4,0,1,1)
Reduced motion	@media (prefers-reduced-motion: reduce): all durations → 0 ms except opacity fades ≤ 100 ms
Z-index	base 0 · sticky 10 · nav 20 · fab 30 · overlay 40 · modal 50 · toast 60
Breakpoints	sm 480 · md 768 · lg 1024 · xl 1280 · 2xl 1536 (min-width)
Layout	content-max 1200 · sidebar 240 · rail 72 · tabbar 64 + safe-area · touch-min 44
Focus	focus-ring 2px solid primary-700, offset 2px (outline, never removed)
Opacity	disabled .45 · scrim rgb(17 24 39 /.48)

18.1 CSS variables and Tailwind mapping

```

:root {
  /* colour - brand */
  --color-primary-50:#ECDFE5; --color-primary-500:#10B981; --color-primary-600:#059669;
  --color-primary-700:#047857; --color-primary-800:#065F46;
  --color-secondary-50:#EEF2FF; --color-secondary-500:#6366F1; --color-secondary-600:#4F46E5;
  --color-accent-50:#FFFBEB; --color-accent-500:#F59E0B; --color-accent-700:#B45309;
  --color-error-50:#FEE2F2; --color-error-500:#EF4444; --color-error-600:#DC2626; --color-error-700:#B91C1C;
  --color-neutral-50:#F9FADF; --color-neutral-100:#F3F4F6; --color-neutral-200:#E5E7EB;
  --color-neutral-300:#D1D5DB; --color-neutral-600:#6B7280; --color-neutral-700:#374151; --color-neutral-900:#111827;
  /* colour - semantic aliases (components use these) */
  --fg-default: var(--color-neutral-900); --fg-muted: var(--color-neutral-600);
  --bg-app: var(--color-neutral-50); --bg-card: #FFFFFF;
  --border-default: var(--color-neutral-200);
  --action-primary-bg: var(--color-primary-700); --action-primary-fg: #FFFFFF;
  --action-danger-bg: var(--color-error-600);
  --money-income: var(--color-primary-700); --money-expense: var(--color-error-700);
  --money-saving: var(--color-secondary-600);
  --status-ontrack-fg: var(--color-primary-800); --status-ontrack-bg: var(--color-primary-50);
  --status-atrisk-fg: var(--color-accent-700); --status-atrisk-bg: var(--color-accent-50);
  --status-behind-fg: var(--color-error-700); --status-behind-bg: var(--color-error-50);
  /* type */
  --font-heading: "Plus Jakarta Sans", system-ui, sans-serif;
  --font-body: "Inter", system-ui, sans-serif;
  /* shape, motion */
  --radius-md: 10px; --radius-lg: 14px; --dur-base: 200ms; --ease-standard: cubic-bezier(.2,0,0,1);
}

/* Tailwind v4 (app/globals.css) */
@import "tailwindcss";
@theme inline {
  --color-primary-700: var(--color-primary-700);
  --color-income: var(--money-income);
  --color-expense: var(--money-expense);
  --color-saving: var(--money-saving);
  --font-heading: var(--font-heading);
  --font-sans: var(--font-body);
  --radius-card: var(--radius-lg);
}

```

18.2 Component rules

- Cards exist only to group related information; no card nested in a card unless the inner item is itself a list item (goal card inside goals list).
- One primary button per view region. Destructive actions never use the primary style.
- Icons: Lucide (rounded outline), 20 px default, 24 px in navigation, stroke 1.75. Category = icon + label, never icon alone.
- Illustrations: flat, soft gradients, people of African descent represented prominently — consistent with the boards and the launch market.

19. Loading, Empty, Error & Offline States

19.1 Global rules

State	Rule
Loading	Skeletons matching final layout for screen content; inline spinners for local actions; never a full-screen spinner once the shell has loaded. Background refresh is silent. Skeleton appears only after 150 ms to avoid flashes.
Empty	Must answer: what is missing, why it matters, what to do next. One primary CTA where appropriate.
Error	Plain language, no codes. Preserve user input. Retry. Validation and auth errors next to the affected field or action. Cached content stays visible with a non-blocking error banner.
Offline	Subtle "Offline" chip in header; creation of transactions and contributions queued; edits/deletes of synced records and goal/couple management disabled with explanation "Connect to the internet to do this."

19.2 State matrix by screen

Screen	Loading	Empty	Error	Offline
Home	Skeleton hero + 3 rows + 2 goal cards	"Add your first income or expense — it unlocks your totals and insights." + Add transaction	Banner over cached data + Retry	Cached summary + queued items applied locally, marked "Sync pending"
Activity	8 skeleton rows	No data: "Nothing recorded yet" + Add. Filtered: "No activity matches these filters" + Clear filters	Retry; keep previous results	Cached list; queued rows at top
Insights	Skeleton cards + chart frames	"Insights improve as you record transactions" + available totals shown	Per-card error with Retry	Last cached period; banner
Goals	3 skeleton cards	My: "You have no savings goals yet" + Create goal. Ours: no partner → Connect partner; partner, no goals → Create shared goal	Retry	Cached list; create disabled
Goal details	Hero skeleton	Zero contributions: \$0 saved, remaining, required pace, Add contribution	Retry	Add contribution queued; status recomputed locally
Couple	Status card skeleton	No partner state (design)	Retry	Read-only
Forms	Submit inline spinner, fields disabled	—	Field errors; top error for server issues; values kept	Save → queued (create only) with toast "Saved offline — will sync"
Auth	Button spinner	—	Generic credential error; lockout message	"You're offline. Connect to sign in."

19.3 Offline architecture (PWA)

1. **Service worker** (Serwist): precache app shell, fonts, icons; runtime cache GET /api/v1 responses with network-first (3 s timeout) then cache.
2. **Read cache**: TanStack Query persisted to IndexedDB (encrypted-at-rest by the OS only; cleared on logout).

3. **Write queue:** IndexedDB store outbox with {id (UUID), endpoint, body, idempotency_key, created_at, attempts}. Client-generated UUID becomes the record id so later edits reference it.
4. **Sync:** triggered on online event, app focus, and every 60 s while items remain (Background Sync API where supported; Safari falls back to these triggers). FIFO, exponential backoff, max 8 attempts.
5. **Conflicts:** creates are idempotent, so retries cannot duplicate. A permanently rejected item (e.g. goal archived meanwhile) moves to a "Needs attention" list with the reason and Edit / Discard actions. Data is never silently dropped.
6. **Logout** with pending items warns: "2 entries haven't synced. Log out anyway and lose them?"

20. Accessibility

Target: **WCAG 2.2 Level AA** for every screen. Accessibility is part of the Definition of Done, not a later pass.

Area	Requirement
Semantics	Landmarks (header, nav, main); one h1 per page; lists as lists; tables as tables; buttons are <button>, links are <a>.
Keyboard	Everything operable by keyboard; visible focus ring (token); logical tab order; skip link; focus trapped in dialogs/sheets and returned on close; no keyboard traps.
Screen readers	Money read naturally ("minus twelve dollars, expense, Food, 17 September"). Status chips announce text. Toasts use role="status"; errors role="alert". Charts expose data tables.
Colour & contrast	Text $\geq 4.5:1$ ($\geq 3:1$ for ≥ 24 px or 18.66 px bold); UI components and focus $\geq 3:1$. Meaning never by colour alone: sign, icon and label accompany every financial colour.
Target size	Interactive targets $\geq 44 \times 44$ px on touch, $\geq 24 \times 24$ px minimum anywhere (WCAG 2.5.8) with spacing.
Text scaling	Works at 200% zoom and 320 px width without horizontal scrolling (reflow).
Forms	Persistent visible labels; hints and errors linked by aria-describedby; errors summarised on submit; autocomplete attributes (name, email, tel, new-password, current-password); no time limits.
Motion	Respect prefers-reduced-motion; no flashing; celebration can be skipped.
Authentication	No cognitive tests; paste allowed in password fields; password managers supported (WCAG 3.3.8).
Language	lang="en" on html; plain language; numbers formatted to locale.
Testing	axe-core in Playwright on every screen state; manual checks with NVDA + Firefox, VoiceOver + Safari (macOS and iOS), TalkBack + Chrome before release.

21. Responsive Behaviour

Mobile-first CSS. Layouts are designed at 360, 768, 1024 and 1440 px and must work continuously between them (no dead zones). Minimum supported width: 320 px.

Screen	< 768 px	768–1023 px	≥ 1024 px
Shell	Bottom tabs + FAB; header 56 px	Icon rail; header	Sidebar 240 px; content max 1200 px centred
Home	Single column: hero → categories → goals	Two columns: hero full width; categories goals	12-col grid: hero (8) + goals (4); categories (8) + recent activity (4)
Add forms	Full-height sheet; Save button fixed above keyboard	Centred dialog 480 px	Centred dialog 480 px
Activity	List; details as pushed page	List; details as pushed page	Split view: list 58% / details 42%
Insights	Stacked cards; charts full width; 2 × 2 metric grid; bar chart shows last 3 buckets	2-column cards	4 metric cards in a row; trend (7) + bars (5); breakdown full width
Goals	One column cards	Two-column grid	Three-column grid
Goal details	Single column; sticky Add contribution bar	Two columns: hero pace/status	Two columns + history full width
Tables in settings	Rows stack label over value	Label left, value right	Label left, value right

- Respect safe-area insets (`env(safe-area-inset-*)`) for the tab bar, FAB and sticky buttons on notched phones.
- Use 100dvh (dynamic viewport) for sheets so the mobile browser toolbar does not hide the Save button; keep the focused input visible when the keyboard opens.
- Hover-only affordances are forbidden; every hover effect has a focus and touch equivalent.
- Images and illustrations use `next/image` with responsive sizes; illustrations hidden below 360 px when they would push content below the fold.

22. Web Acceptance Criteria

A criterion passes only when an automated test exists and passes (Section 23), unless marked "manual".

ID	Criterion	Traces to
WAC-01	A new user can register with email or phone, verify, set base currency and reach Home in under 90 seconds; duplicate identifiers are rejected without revealing account existence.	AC01, FR-01, FR-05
WAC-02	Login sessions survive reloads and browser restarts; logout clears cookies, query cache and offline stores.	AC01, FR-02/03
WAC-03	Creating, editing and deleting income updates Home, Activity and Insights totals for all affected periods without a manual refresh.	AC02
WAC-04	Same as WAC-03 for expenses, including category totals and percentages.	AC03
WAC-05	Home shows income, expenses, recorded savings, remaining cash flow, net cash flow and savings rate matching F-01...F-06 for Today, This week and This month.	AC04
WAC-06	A period with zero income shows savings rate "N/A" and no errors anywhere in UI, API or logs.	AC05
WAC-07	Individual goals can be created with name, target amount, currency and target date; invalid values are blocked inline.	AC06
WAC-08	Users in an active couple can create couple goals; users without one see the option disabled with an explanation and receive 409 COUPLE_REQUIRED from the API.	AC07
WAC-09	Goal balance always equals the sum of its contributions after any create, edit or delete.	AC08
WAC-10	Progress, remaining amount, required pace, status and projection match F-11...F-20 for the reference cases in Section 23.	AC09
WAC-11	A goal becomes Completed the moment contributions reach the target, and reverts if a contribution is removed.	AC10
WAC-12	Both partners see every couple-goal contribution with contributor name; no other partner data is visible in UI or obtainable via API (verified by RLS tests).	AC11, AC12
WAC-13	Daily, weekly and monthly insights calculate correctly across timezone and month boundaries and show previous-period comparison when data exists.	AC13
WAC-14	An amount entered in a non-base currency is stored with its original value and converted at the rate of its date; the preview matches the saved value.	Scope exception
WAC-15	Changing base currency re-expresses all totals and preserves original amounts.	Scope exception
WAC-16	Core flows pass E2E on Chrome, Safari, Firefox, Edge (latest 2) and on iOS Safari and Android Chrome viewports 360–430 px.	AC14
WAC-17	With the network off, a user can add an expense; it syncs once online with no duplicate.	Design §21
WAC-18	Every primary data screen has loading, empty and error states as specified in 19.2.	Design AC8
WAC-19	No axe-core violations of serious or critical impact; manual screen reader pass completed (manual).	Design §23
WAC-20	Home loads with LCP ≤ 2.5 s on a mid-range phone over 4G (Lighthouse mobile profile); dashboard API p95 ≤ 800 ms server time.	PRD §12

23. Testing Architecture

Layer	Tool	Scope	Gate
Unit (domain)	Vitest	Every formula F-01...F-24, status, rounding, date/period boundaries	100% line + branch coverage for packages/domain; blocks merge
Schema & RLS	pgTAP via supabase test db	Constraints, triggers, functions, and every RLS policy for owner / partner / stranger / ex-partner	Blocks merge
API integration	Vitest + local Supabase (Docker)	Each endpoint: happy path, validation, auth, 404 isolation, idempotency	Blocks merge
Component	Vitest + Testing Library	Forms, AmountInput, pickers, states	Blocks merge
E2E	Playwright (Chromium, WebKit, Firefox; Pixel 7 + iPhone 14 emulation)	Flows 7.1-7.10, offline sync, couple two-user scenario	Blocks deploy to production
Accessibility	@axe-core/playwright + manual	All screens and states	Serious/critical = fail
Performance	Lighthouse CI + k6 (API)	Budgets in WAC-20; JS ≤ 180 kB gzip on first load of /home	Warn on preview, fail on main
Visual	Playwright screenshots	Key screens at 360 / 768 / 1280	Reviewed diffs

23.1 Required domain test cases (PRD §12)

ID	Case	Expected
T-01	Totals: income 1,200.00, expenses 570.00, savings 300.00	net 630.00, remaining 330.00, rate 25.0%
T-02	Zero income, expenses 50.00	rate N/A; remaining -50.00; no exception
T-03	Contribution 100.00 in same period	expenses unchanged; saved +100.00
T-04	Goal target 1,200, balance 600	progress 50%, remaining 600
T-05	Balance 1,300 on target 1,200	progress 100%, remaining 0, COMPLETED
T-06	Target date today, remaining 60	required daily 60 (no divide-by-zero)
T-07	Target date yesterday, not complete	status BEHIND, required pace overdue
T-08	Couple goal: A 480, B 320 on target 2,000	balance 800, shares 60% / 40%
T-09	Status thresholds at ratios 0.95, 0.9499, 0.75, 0.7499	ON_TRACK, AT_RISK, AT_RISK, BEHIND
T-10	New goal, zero contributions, day 0	ON_TRACK
T-11	Expense 5,000 LRD with rate 1 USD = 189.39 LRD	base 26.40 USD (half away from zero)
T-12	JPY 1,000 → USD, KWD 1.234 → USD	correct exponents 0 and 3
T-13	Expense at 23:30 on 31 Aug in Africa/Monrovia vs Asia/Tokyo user	falls in the correct local month
T-14	Week boundaries: Sunday vs Monday	ISO week Monday start
T-15	Avg daily spending on day 17 of current month vs full previous month	divide by 17 vs 31
T-16	Previous period = 0	change shown as "New"

23.2 Critical RLS tests

- Partner cannot SELECT, UPDATE or DELETE the other partner's transactions, individual goals or contributions to individual goals (expect 0 rows / error).
- Stranger cannot read any couple goal or contribution by id (404 at API, 0 rows at DB).
- Ex-partner can read archived couple goal history but cannot insert contributions.
- User cannot insert a contribution with user_id of another user; cannot create a couple goal for a couple they are not in.
- accept_invitation rejects: wrong token, expired, invitee already coupled, invitee identity mismatch, reuse of accepted token.

24. Deployment Architecture

24.1 Environments

Environment	Web	Database	Purpose
Local	next dev	Supabase CLI (Docker) with seed	Development and all automated tests
Preview	Vercel preview per pull request	Staging Supabase project	Review, E2E, Lighthouse
Staging	Vercel (main branch, protected)	Staging Supabase project	Release candidate, manual QA
Production	See 24.2	Production Supabase project	Users

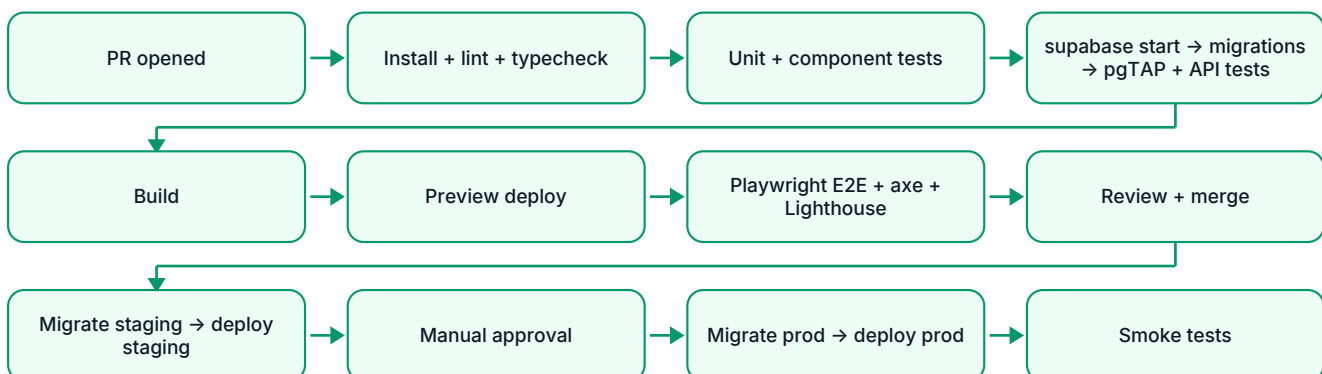
24.2 Hosting phases and cost

Phase	Web hosting	Supabase	Monthly cost	Constraints
1. Build & demo	Vercel Hobby	Free plan (2 projects: staging + demo)	\$0	Hobby is for personal, non-commercial use only. Free projects pause after 7 days without activity and have no backups. Acceptable only before real users.
2. Launch (recommended)	Hetzner VPS (Docker, Nginx, PM2 or Node standalone output, Certbot) or Vercel Pro	Pro plan	Supabase Pro \$25 + VPS (≈ single-digit USD) or Vercel Pro \$20/seat	Backups, no pausing. VPS option needs the hardening checklist and uptime monitoring.

Warning

Do not put paying or real users on Phase 1 infrastructure. A financial app without database backups is a data-loss incident waiting to happen.

24.3 CI/CD pipeline (GitHub Actions)



- Migrations run with `supabase db push` using a deploy token; they must be backward compatible with the currently running app (expand → migrate → contract).
- Generated DB types are committed; CI fails if they are out of date.
- Rollback: redeploy previous web build instantly; database fixes are forward-only migrations.
- Secrets live in GitHub Environments and the host's env store: `SUPABASE_URL`, `SUPABASE_ANON_KEY`, `SUPABASE_SERVICE_ROLE_KEY` (server only), `FX_FALLBACK_KEY`, `SENTRY_DSN`, SMTP/SMS credentials. The service-role key is never exposed with a `NEXT_PUBLIC_` prefix.

24.4 Operations

Concern	Implementation
Monitoring	Sentry (web + API errors, performance traces); uptime check on /api/health every 5 min; Supabase logs.
Scheduled jobs	pg_cron: fx-sync (daily), idempotency purge (hourly), soft-delete purge (daily, > 30 days), invitation expiry (hourly).
Backups	Supabase Pro daily backups; weekly logical dump (pg_dump) encrypted to Cloudflare R2 with 8-week retention; quarterly restore drill.
Security headers	Strict CSP (nonce-based scripts), HSTS, X-Content-Type-Options, Referrer-Policy strict-origin-when-cross-origin, Permissions-Policy, frame-ancestors none.
Email / SMS	Supabase Auth emails via custom SMTP (Resend) on a verified domain (SPF, DKIM, DMARC). SMS via Twilio for phone accounts and phone invitations — this is a per-message cost.
Data protection	HTTPS only; encryption at rest (Supabase); logs and analytics exclude amounts, notes and category names.

25. Security & Success Metrics

25.1 Security checklist

- Passwords hashed by Supabase Auth (bcrypt); minimum 10 characters; breached-password check enabled.
- Sessions in HTTP-only, Secure, SameSite=Lax cookies; no tokens in localStorage (PRD §8).
- Server-side validation (Zod) on every write; amounts positive integers; dates $\leq$ today; notes length-capped and rendered as text (no HTML).
- Authorization at API guard + RLS; 404 for foreign resources; invitation tokens 32 random bytes, stored hashed, single use, 7-day expiry.
- Rate limiting on auth and writes; generic auth errors; lockout messaging without enumeration.
- Dependency scanning (Dependabot) and secret scanning enabled on the repository.

25.2 Success metrics (PRD §13)

Category	Metric
Activation	Account created → base currency set → first transaction → first goal → partner connected (where applicable).
Engagement	Transactions per active user per week; contributions per active user per month; insight views.
Retention	Users still recording transactions after 7 and 30 days.
Outcome	% of active users with an active goal; % reporting clearer understanding of spending and savings progress (in-app one-question survey after 30 days).

26. Known Limitations (Excluded by Scope)

The owner chose strict PRD scope. These gaps are intentional in the MVP and should be reviewed for version 1.1:

Limitation	Consequence in MVP
No withdrawals from goals	Money used from savings cannot be recorded against a goal; users would have to delete or edit contributions, which rewrites history.
No recurring income/expenses	Salary, rent and subscriptions are entered manually each time.
Minimal couple dissolution	Ending a couple archives shared goals read-only; no splitting of shared balances.
No budgets, bank sync, receipts	Per PRD non-goals.
Light theme only	Dark mode not specified; tokens are structured so it can be added without refactoring.
Rates are daily	Conversions use the daily rate of the record's date, not intraday or parallel-market rates.

27. Definition of Done

The MVP is complete when all of the following are true in production:

1. Authentication (register, verify, login, logout, reset), base-currency setup and session persistence work on all supported browsers.
2. Income and expense tracking with multi-currency conversion, editing, deletion with undo, and correct recalculation of every summary.
3. Period calculations for today/week/month and daily/weekly/monthly insights with previous-period comparison.
4. Individual goals, couple invitations, couple goals, contributions, derived balances, required pace, status and projection.
5. Couple privacy guaranteed by RLS and proven by automated tests.
6. All WAC-01...WAC-20 pass; domain coverage 100%; no serious/critical accessibility violations.
7. Loading, empty, error and offline states implemented for every primary screen.
8. Phase 2 infrastructure live: Supabase Pro with backups, monitoring, alerting, security headers and a completed restore drill.